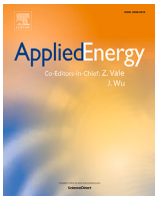
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Coordinating data centre microgrids and mobile power sources for distribution network resilience: A data-driven distributionally robust optimisation approach

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HIGHLIGHTS

- A coordination strategy is proposed for data centre microgrids and mobile power sources.
- An asymmetric data-driven robust optimisation model is developed to address uncertainties.
- A problem-tailored recourse reformulation is established to improve model tractability.
- Dynamic network reconfiguration with detailed load energisation is incorporated.

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ABSTRACT

High-impact and low-probability events are arising more frequently than ever before and pose a great challenge to power system resilience. As an emerging class of critical infrastructure, data centres (DCs) have become increasingly critical for supporting digital services, cloud computing, and intelligent energy applications, and their energy supply should be prioritised during extreme events. Except for their role as critical loads, DCs can also be managed as data centre microgrids (DCMGs) and contribute to distribution network resilience via flexible workload migration. However, the resilience perspective of DCMGs and their coordination with upper-level restoration resources have not been fully investigated. Meanwhile, mobile power sources (MPSs), including mobile energy storage systems and mobile emergency generators, have been increasingly deployed in distribution networks owing to their flexibility and mobility in supporting post-disaster load restoration. The resilience-oriented coordination between DCMGs and MPSs remains insufficiently addressed in the existing literature. To fill this gap, this paper develops a resilience-oriented coordination framework for DCMGs and MPSs to enhance effective load restoration in distribution networks. The proposed framework co-optimises system restoration, MPS dispatch, and DC operations, including workload migration and thermal dynamics. To address uncertainties associated with renewable energy sources, load profiles, and outdoor temperature around DCMGs, a two-stage robust optimisation model is formulated with an asymmetric data-driven risk-tunable uncertainty characterisation. This uncertainty set distinguishes the unequal operational impacts of downward renewable generation deviations, demand increases, and temperature rises from those of their opposite directions. The resulting two-stage robust model is reformulated with a topology-cleaned linear recourse problem and solved using a column-and-constraint generation algorithm. Comprehensive case studies on three distribution networks with different numbers of DCMGs demonstrate the effectiveness of the proposed approach in coordinating different types of resources, supporting load restoration, and enhancing robustness against asymmetric uncertainties.

1. Introduction

Extreme weather events, such as hurricanes, storms, and earthquakes, are low-probability but high-impact hazards that can severely disrupt power systems and cause widespread outages [1]. During such disruptions, a key objective of resilient power systems is to maintain supply continuity for essential loads, including data centres (DCs),

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0306-2619/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Nomenclature

A. Indices and Sets

$i \in \mathcal{DC}$ Index and set of data centre microgrids (DCMGs)
$t \in T$ Index and set of scheduling periods
$s \in \mathcal{S}$ Index and set of server operating states
$i \in \mathcal{MPS}$ Index and set of mobile power sources
$g \in \mathcal{MEG}$ Index and set of mobile emergency generators (MEGs)
$k \in \mathcal{MES}$ Index and set of mobile energy storage systems (MESSs)
$n \in N_{ms}$ Index and set of MPS connection nodes
$b \in \mathcal{B}$ Index and set of buses
$(b,p) \in \mathcal{L}$ Index and set of distribution lines
$d \in \mathcal{D}$ Index and set of conventional loads
$g \in \mathcal{DG}$ Index and set of conventional diesel generators (DGs)
$g \in \mathcal{PV}$ Index and set of photovoltaic units

B. Parameters

$W_{i,t}^{0}$ Original workload demand of DC i at time step t
$\overline{F}_{ij}$ Maximum workload migration capacity from DC i to DC j
$\overline{F}_{i}^{\text{out}}$ Maximum outgoing migrated workload from DC i
$\overline{N}_{i}$ Maximum number of servers in DC i
$\mu_{i,s}$ Service rate of a server at state s in DC i
T_{i}^{delay} Maximum allowable average response delay of DC i
$P_{i,s}^{\text{fix}}$ Fixed server power at state s in DC i
$\alpha_{i,s}$ Dynamic server power coefficient at state s in DC i
$\overline{P}_{i}^{\text{col}}$ Maximum cooling power of DC i
R_i, C_i Thermal resistance/capacitance of DC i
$\eta_i^{\text{h}}, \eta_i^{\text{c}}$ Server-to-heat conversion coefficient/cooling coefficient of performance (COP) of DC i
$\underline{Tem}_{i}^{\text{in}}, \overline{Tem}_{i}^{\text{in}}$ Minimum/maximum indoor temperature of DCMG i
$\underline{P}_{g}^{eg}, \overline{P}_{g}^{eg}$ Minimum/maximum active power output of MEG g
$\underline{Q}_{g}^{eg}, \overline{Q}_{g}^{eg}$ Minimum/maximum reactive power output of MEG g
$\overline{P}_{k}^{es}, \overline{E}_{k}^{es}$ Power/energy capacity of MESS k
$\underline{S}_{k}^{es}, \overline{S}_{k}^{es}$ Minimum/maximum state of charge of MESS k
$\eta_k^{esc}, \eta_k^{esd}$ Charging/discharging efficiency of MESS k
c_d^{ls} Load shedding cost of load d
$\underline{P}_{g}^{dg}, \overline{P}_{g}^{dg}$ Minimum/maximum active power output of DG g
$\underline{Q}_{g}^{dg}, \overline{Q}_{g}^{dg}$ Minimum/maximum reactive power output of DG g
$\tilde{P}_{g,t}^{pv}$ Available power of PV g at time step t
$\overline{S}_{g}^{pv}$ Apparent power capacity of PV unit g
$\overline{P}_{d,t}^{ed}, \overline{Q}_{d,t}^{ed}$ Active/reactive load d at time step t
$\underline{V}, \overline{V}$ Lower/upper voltage limits
$\overline{S}_{bp}, \overline{F}_{bp}$ Thermal limit/virtual power flow limit of line (b,p)
r_{bp}, x_{bp} Resistance and reactance of line (b,p)

C. Variables

$W_{i,t}$ Processed workload of DCMG i at time step t
$F_{ij,t}$ Workload migrated from DCMG i to DCMG j at time step t
$N_{i,s,t}$ Number of active servers at state s in DCMG i at time step t
$P_{i,t}^{\text{svr}}, P_{i,t}^{\text{col}}$ Server/cooling power consumption of DCMG i at time step t
$Tem_{i,t}^{\text{in}}, Tem_{i,t}^{\text{out}}$ Indoor/outdoor temperature of DCMG i at time step t
$P_{i,t}^{\text{dc}}$ Total electrical demand of DCMG i at time step t
$u_{i,n,t}$ Binary routing variable of MPS i at node n and time step t
$P_{g,n,t}^{eg}, Q_{g,n,t}^{eg}$ Active/reactive power output of MEG g at node n and time step t
$P_{k,n,t}^{esc}, P_{k,n,t}^{esd}$ Charging/discharging power of MESS k at node n and time step t
$u_{k,t}^{es}$ Binary charging/discharging status of MESS k at time step t
$S_{k,t}^{es}$ State of charge of MESS k at time step t
$P_{d,t}^{ed}, Q_{d,t}^{ed}$ Restored active/reactive load of load d at time step t
$P_{g,t}^{dg}, Q_{g,t}^{dg}$ Active/reactive power output of DG g at time step t
$P_{g,t}^{pv}, Q_{g,t}^{pv}$ Active/reactive power output of PV unit g at time step t
$e_{b,t}, y_{bp,t}$ Binary energisation status of bus b/line (b,p) at time step t
$V_{b,t}$ Voltage magnitude of bus b at time step t
$P_{bp,t}, Q_{bp,t}$ Active/reactive power flow on line (b,p) at time step t
$F_{a,t}^{s}$ Virtual power injection of black-start source a at time step t
$F_{bp,t}$ Virtual power flow on line (b,p) at time step t

medical facilities, and police stations, which gives rise to the system load restoration problem [2]. Among these critical loads, DCs are becoming increasingly important in the digital and artificial intelligence (AI) era. Their electricity consumption accounts for approximately 1.5% of global electricity demand, and this share is expected to increase to 3–4% by 2030 [3]. Meanwhile, power supply interruptions have been responsible for more than 25% of reported DC outages in the United States, resulting in substantial economic losses [4]. Therefore, as AI and cloud services continue to expand, enhancing the power resilience and sustainable operation of DCs has become an urgent priority for both critical infrastructure protection and global decarbonisation.

In this context, a data centre microgrid (DCMG) can be viewed as a microgrid that supplies one or multiple data centres while coordinating on-site generation, energy storage, cooling facilities, and information-processing loads [5]. Unlike conventional industrial microgrids, whose operational flexibility is generally constrained by production schedules, material flows, and safety requirements, DCMGs exhibit distinctive cyber-physical flexibility arising from the coupling among electrical operation, computing workloads, server states, and thermal dynamics. In particular, computing workloads can be transferred among geographically distributed DCMGs subject to server-capacity and quality-of-service (QoS) constraints, thereby reshaping the spatial distribution of electricity demand without installing additional generation or storage resources [5,6]. Therefore, in addition to being critical loads requiring a reliable power supply, DCMGs can actively support post-disaster restoration through workload migration and coordinated interactions with distributed energy resources (DERs). Meanwhile, mobile power sources (MPSs), as an emerging type of DER, have been applied in system load restoration owing to their mobility and operational flexibility [7]. Coordinating the spatial demand-side flexibility of DCMGs with the mobile supply-side flexibility of MPSs thus provides a promising pathway for enhancing power system resilience during the post-disaster period.

1.1. Data centre microgrids for resilience

Microgrids (MGs) have been widely recognised as an effective solution for enhancing power system resilience against extreme weather events [1]. By integrating DERs, energy storage systems (ESSs), and local control capabilities [8], MGs can operate in both grid-connected and islanded modes, thereby maintaining power supply to critical loads when the main grid is partially or fully unavailable. In Wang et al. [9], a two-stage optimisation framework is proposed to enable the evaluation of MGs' restoration actions and their feasibility via an unbalanced three-phase optimal power flow. In Qiu et al. [10], a robust dual dynamic integer programming approach is proposed to efficiently tackle the multi-stage resilient scheduling problem with the objective of maximising load restoration during tropical cyclones. In Hussain et al. [11], a consensus-based distributed algorithm is proposed to schedule dispatchable DERs with the objective of maximising the supply adequacy of MGs. In Shen et al. [12], a distributed coordination approach based on the alternating direction method of multipliers is proposed to make

load restoration decisions for MGs, while ensuring that the risk-limiting constraints of the overall system are satisfied.

Existing studies have shown that the ability of MGs to provide localised energy support makes them highly valuable in post-disaster service recovery [13], especially for infrastructures requiring uninterrupted or prioritised power supply, such as DCs [14]. In this context, DCs have attracted growing attention from a resilience perspective, leading to a growing body of research that considers DCs as critical loads in resilience studies. For instance, in Liu et al. [15], an approach is proposed for post-event recovery of distribution networks, treating DCs as the critical load to minimise utility losses during the recovery stage. In Tao et al. [16], a co-optimisation model is proposed for integrated urban energy systems to improve resilience during extreme events, considering DCs' optimal operation, workload redistribution, and waste heat reuse.

Going further, existing studies generally treat DCs as critical loads and focus on managing MGs to restore power to these DCs during extreme weather events. Recent research highlights that DCs should not be regarded merely as uniform electricity-demand blocks and that their internal computational flexibility should be made interpretable to energy systems [17]. This perspective provides an important conceptual foundation for exploiting workload migration to reshape the spatial distribution of electricity demand during system restoration. Nevertheless, existing studies do not leverage the coordination of DCMGs as load restoration resources to enhance upper-level distribution network resilience. While several studies have explored the capability and coordination of MGs for enhancing upper-level network resilience [18,19], none incorporates the operational flexibility of DCs into their coordination schemes. More specifically, most research focuses on isolated or self-contained DCMG operations within energy management contexts, with limited attention to coordinated resilience support between multiple DCMGs. Furthermore, various emergency resources, such as MPSs, have been widely studied for post-disaster load restoration. Investigating the coordination between DCMGs and these advanced emergency resources remains an open area that has yet to be addressed in the literature.

1.2. Mobile power sources for resilience

In the existing literature, the resilience-oriented dispatch of MPSs has been widely studied. On the one hand, in Kumari and Ganguly [20], the routing and scheduling behaviours of mobile emergency generators (MEGs) are simultaneously optimised to supply islanded sections and enhance distribution network resilience. In Zhang et al. [21], a three-stage stochastic optimisation model is proposed for MEG allocation in resilient distribution networks, considering the planning stage, preventive response stage, and emergency response stage. In Lei et al. [22], MEGs are utilised as critical flexibility resources of distribution networks for resilient emergency response to extreme weather events, where both the pre-positioning and dispatch of MEGs are incorporated into the resilient framework. In Cai et al. [23], a two-stage dispatching model is proposed for MEGs to reduce system outage duration and sequentially restore out-of-service loads. In Wu et al. [24], MEGs are deployed in the post-disaster restoration of transmission and distribution systems, where MEGs are dispatched to restart non-black-start generators or support critical loads. On the other hand, in Rajabzadeh and Kalantar [25], mobile energy storage systems (MESSs) are deployed to restore critical loads for resilience enhancement of distribution networks. In Ghasemi and Moshtagh [26], a load restoration strategy based on the real-time dispatch of MESSs is proposed for resilience enhancement, considering both black-start and non-black-start distributed generators. In Chen et al. [27], a two-stage stochastic model is developed for the dispatch of MESSs, considering various operating scenarios such as normal, severe, and extreme scenarios. In Mishra et al. [18], the coordination between MESSs and multiple microgrids is formulated in a resilience quantification framework to assess the resilience of a distribution network.

It is worth noting that prompt load restoration can be influenced by both the operability of MEGs and MESSs, while effective coordination between different types of MPSs can significantly enhance the speed and quality of service restoration [28]. It is difficult to achieve the optimal load restoration solely based on a single type of MPS, leading to the requirement for integrated methods that can coordinate available flexibility resources. In this context, there have been a few studies focusing on the resilience-oriented coordination problem of MEGs and MESSs. For example, in Lei et al. [29], a two-stage robust optimisation framework is proposed for the resilience-oriented coordination between MEGs and MESSs in distribution networks. In Wang et al. [30], a model-free and multi-agent learning framework is proposed for the coordination of different types of MPSs, including MEGs and MESSs. In Yang et al. [31], a two-stage restoration scheme considering both MEGs and MESSs is proposed to facilitate the distribution network restoration following seismic disasters. In Erenoğlu and Erdinç [32], a real-time optimisation-based critical load restoration strategy is proposed for the post-event stages, considering the routing and scheduling behaviours of MEGs and MESSs. However, all the above works only consider the benefits of MPSs for resilience enhancement, without considering the coordination between MPSs and emerging DCMGs, despite their capability for flexible load migration, fast islanded operation, coordinated energy sharing, and highly reliable local backup power support for critical loads.

1.3. Uncertainty modelling

Effective uncertainty management is a critical challenge in resilience-oriented system operation and control, particularly in distribution networks with high penetration of RESs. Existing uncertainty modelling approaches can be broadly categorised into stochastic programming (SP) and robust optimisation (RO) [33]. On the one hand, SP represents uncertainty through multiple scenarios and seeks decisions that perform well over these realisations [34]. Although suitable for scenario-based modelling, SP generally requires a large amount of data to accurately estimate the underlying probability distribution, which may significantly increase the computational burden. On the other hand, RO focuses on worst-case realisations to guarantee solution robustness under uncertainty. However, such solutions may become overly conservative, leading to higher operating costs and reduced efficiency. Moreover, the construction of RO models often requires prior knowledge of the uncertainty bounds [35], which may not be available. In the existing literature on resilience-oriented power system operations, both SP and RO have been widely adopted to characterise uncertainties. Specifically, SP has been used to model uncertainties in RES generation and demand, while RO has mainly been applied to address worst-case scenarios under extreme events [1].

In this context, distributionally robust optimisation (DRO) provides a more flexible alternative by describing uncertainty through ambiguity sets, thereby reducing the dependence on exact distributional assumptions and offering a better balance between the conservatism of RO and the data dependence of SP. Nevertheless, only a limited number of studies have applied DRO to DC-related optimisation problems. For example, in Li et al. [36], a distributionally robust chance-constrained model based only on first- and second-order moment information is proposed, neglecting richer statistical features and resulting in a complicated reformulation. In Wang et al.,Han et al. [37,38], the computational burden is reduced through ambiguity sets based on the 1-norm or ∞-norm distances; nevertheless, their formulations remain sensitive to the number of involved scenarios. In Duan et al. [39], conventional Wasserstein-based DRO is employed to directly embed the ambiguity set into the optimisation model through reformulated distributionally robust chance constraints. However, this setting causes the problem complexity to increase with the dimension and structure of uncertain parameters.

To overcome this limitation, in Zheng et al. [40], a data-driven risk-tunable bound is derived from the Wasserstein ambiguity set, reformulating the original distributionally robust chance-constrained problem into a tractable risk-bound optimisation model, thereby reducing modelling complexity while preserving the data-driven nature

of DRO. However, the uncertainty treatment in Zheng et al. [40] still has limitations when extended to the present resilience-oriented coordination problem. First, its risk-tunable uncertainty characterisation is essentially channel-wise and symmetric, which is not sufficiently expressive for resilience applications where PV downward deviations, load increases, and temperature rises usually have significantly different operational impacts from the opposite directions. Second, the formulation in Zheng et al. [40] is developed for islanded DCMG operation, whereas the present work considers a much more tightly coupled restoration setting involving MPSs, network reconfiguration, and post-disaster load restoration. Third, directly applying its uncertainty treatment to the integrated resilience model would still face tractability challenges due to the enlarged problem scale and the stronger coupling between discrete restoration decisions and continuous recourse actions.

1.4. Research contributions

This paper aims to address the above challenge by introducing a resilience-oriented coordination framework considering both DCMGs and MPSs in distribution networks, where uncertainties associated with RESs, demand, and outdoor temperature are realised via an asymmetric data-driven risk-tunable characterisation based on the Wasserstein distance. More specifically, the novel contributions of this paper are described as follows:

1. A coordination strategy considering both DCMGs and MPSs, including MEGs and MESSs, is proposed for the resilience enhancement of the distribution network, where the load migration capability of DCMGs as well as the routing and scheduling behaviours of MPSs is explicitly modelled.
2. A two-stage robust optimisation model is developed to address uncertainties in RESs, demand, and outdoor temperature. An asymmetric data-driven risk-tunable uncertainty characterisation based on the Wasserstein distance is introduced to distinguish the unequal operational impacts of downward RES deviations, demand increases, and temperature rises from those in the opposite directions. The resulting formulation derives explicit uncertainty bounds from historical data without relying on detailed parametric assumptions, while preserving tractability.
3. A problem-tailored recourse reformulation is established to improve the tractability of the resulting large-scale mixed-integer robust model. After the first-stage decisions are fixed, redundant topology-construction constraints and auxiliary variables are removed, while the relevant quadratic constraints are polygonally linearised. Consequently, the second-stage problem is reduced to an LP that permits valid linear dualisation and is solved using a column-and-constraint generation (C&CG) algorithm.
4. Dynamic network reconfiguration with detailed load energisation is incorporated to more realistically model post-disaster load restoration. In addition, the black-start capability of grid-forming PV units is explicitly considered, allowing renewable resources to actively support restoration in the early recovery stage.

1.5. Paper organisation

The rest of this paper is organised as follows. Section 2 outlines the proposed resilience-oriented coordination framework. Section 3 presents the detailed mathematical formulation of the proposed optimisation model, including the DCMG operation model, the MPS routing and scheduling model, and the power network operation model. Section 4 introduces the developed asymmetric data-driven framework with scenario accumulation and topology-cleaned recourse for uncertainty characterisation and a robust solution. Section 5 describes the experimental setup, while Section 6 reports and analyses the numerical results. Finally, Section 7 concludes this paper.

2. Outline of the proposed resilience-oriented coordination framework

This paper proposes a resilience-oriented coordination framework that jointly considers DCMGs and MPSs, including MEGs and MESSs. On the DCMG side, critical computing workloads can be protected through workload migration. Specifically, a DCMG located in an outage-affected area can migrate part of its critical workload to another DCMG outside the disrupted area, thereby maintaining service continuity. Such workload migration also reduces the local electricity demand in the outage area and alleviates the supply pressure during post-disaster restoration. In addition, when sufficient on-site generation or storage capacity is available, DCMGs can interact with the upstream distribution network and provide flexible power support for system-level resilience enhancement. On the MPS side, MEGs and MESSs can route through the road network and be connected to selected buses in the distribution network. By optimising their routing, connection locations, and charging/discharging or generation schedules, MPSs can provide temporary power support to damaged areas and assist the restoration of critical loads (Fig. 1).

In this context, the coordination between DCMGs and MPSs is essential because their resilience contributions are strongly coupled. On the one hand, workload migration among DCMGs reshapes the spatial distribution of electricity demand, which affects where mobile power support is most needed. On the other hand, the deployment of MPSs can relieve the power supply pressure faced by DCMGs and the surrounding distribution network. In particular, MPSs can either support the upper-level distribution network to improve overall load restoration, or directly assist a local DCMG by connecting to the same bus or nearby buses and supplying power to its critical computing workloads. Therefore, resilience-oriented coordinated optimisation of DCMG workload migration as well as MPS routing and scheduling is required to fully exploit their complementary flexibility during the post-disaster period.

3. Mathematical formulations of the proposed coordination framework

The proposed resilience-oriented coordination framework for DCMGs and MPSs explicitly incorporates the workload migration capability of DCs, along with the routing and scheduling behaviours of MPSs. The overall formulation is organised into three interconnected components: 1) the DCMG operation model, 2) the MPS routing and scheduling model, and 3) the power network model.

3.1. Data centre microgrid operation model

The DCMG operation model is formulated by jointly considering workload migration, server dispatch, service quality, and thermal-cooling coupling [40]. These components collectively characterise the interactions between cyber resources (computing workloads and servers) and physical resources (power and cooling systems) within each DCMG. Let $i \in DC$ denote the index of a DCMG, $t \in \mathcal{T}$ is the scheduling interval and $s \in S$ corresponds to the server operating state.

3.1.1. Workload allocation and migration

Workload allocation across geographically distributed DCs relies on a communication network. Let $W_{i,t}$ denote the workload processed by DCMG i at time step t, and let $TF_{ij,t}$ denote the workload migrated from DCMG i to DCMG j at time t. The workload balance is given by

$$W_{i,t} = W_{i,t}^{0} + \sum_{j \in DC, j \neq i} TF_{ji,t} - \sum_{j \in DC, j \neq i} TF_{ij,t}, \forall i \in DC, \forall t \in T \tag{1}$$

where $W_{i,t}^{0}$ is the original workload demand of DCMG i. The migrated workload is limited by the communication bandwidth between DCMGs:

$$0 \leq TF_{ij,t} \leq z_{ij,t}\overline{TF}_{ij}, \forall i, j \in DC, i \neq j, \forall t \in T \tag{2}$$

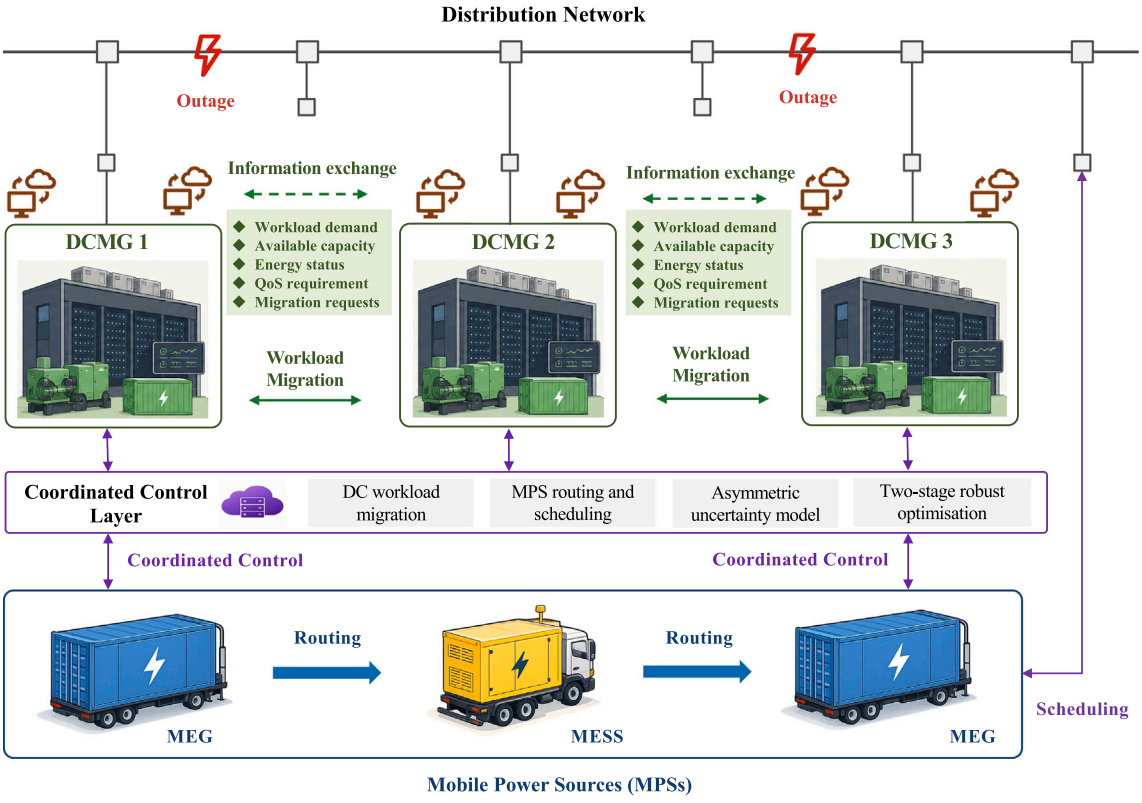


Fig. 1. Illustration of the coordination between DCMGs and MPSs towards enhancing the resilience of a distribution network after natural disasters.

$$z_{ij,t} + z_{ji,t} \le 1, \forall i, j \in DC, i \ne j, \forall t \in T \tag{3}$$

where $z_{ij,t}$ is a binary variable introduced to prevent simultaneous bidirectional workload migration between two DCMGs.

To reflect the limited communication and service-transfer capability of each DCMG, the total outgoing migrated workload is constrained by

$$\sum_{j \in DC, j \ne i} TF_{ij,t} \le \overline{TF}_i^{\text{out}}, \forall i \in DC, \forall t \in T \tag{4}$$

where $\overline{TF}_i^{\text{out}}$ is the maximum outgoing workload transfer capacity of DCMG i.

3.1.2. Server-state dispatch

Let $N_{i,s,t}$ denote the number of active servers operating at state s in DCMG i at time t. Then, the number of active servers cannot exceed the installed capacity:

$$\sum_{s \in S} N_{i,s,t} \le \overline{N}_i, \forall i \in DC, \forall t \in T \tag{5}$$

where $\overline{N}_i$ is the maximum number of servers in DCMG i.

3.1.3. QoS constraint

Let $\mu_{i,s}$ denote the service rate of a server at state s in DCMG i, and let T_i^{delay} denote the maximum allowable average response delay. Following the queueing-based approximation adopted in Zheng et al. [40], the quality-of-service (QoS) requirement is enforced as

$$\sum_{s \in S} \mu_{i,s} N_{i,s,t} - W_{i,t} \ge \frac{1}{T_i^{delay}}, \forall i \in DC, \forall t \in T \tag{6}$$

3.1.4. Server power consumption

The server power consumption of a DC consists of fixed and dynamic components. The fixed component represents the baseline power required to keep servers operating in different states, while the dynamic component is associated with the processor operating frequency and can be adjusted through dynamic voltage and frequency scaling (DVFS) [16]. In detail, the server power consumption can be formulated as

$$P_{i,t}^{\text{svr}} = P_{i,t}^{\text{fix}} + P_{i,t}^{\text{dyn}}, \forall i \in DC, \forall t \in T \tag{7}$$

$$P_{i,t}^{\text{fix}} = \sum_{s \in S_i} N_{i,s,t} P_{i,s}^{\text{fix}}, \forall i \in DC, \forall t \in T \tag{8}$$

$$P_{i,t}^{\text{dyn}} = \sum_{s \in S_i} N_{i,s,t} k_{i,s} f_{i,s}^3, \forall i \in DC, \forall t \in T \tag{9}$$

where $P_{i,t}^{\text{svr}}$ denotes the total server power consumption of DCMG i at time t, and $N_{i,s,t}$ is the number of servers operating in state s. $P_{i,s}^{\text{fix}}$ represents the fixed power consumption of one server in state s, while $k_{i,s}$ corresponds to the dynamic power coefficient and $f_{i,s}$ is the processor frequency of state s.

3.1.5. Thermal dynamics

Let $P_{i,t}^{\text{col}}$ denote the cooling power consumption of DC i at time t. The cooling system is bounded by its installed capacity:

$$0 \le P_{i,t}^{\text{col}} \le \overline{P}_i^{\text{col}}, \forall i \in DC, \forall t \in T \tag{10}$$

where $\overline{P}_i^{\text{col}}$ is the maximum cooling power.

The thermodynamic process in a DC can be modelled by a first-order linear differential equation derived from the dynamic heat balance, using an equivalent thermal parameter model and discretised under the quasi-steady-state assumption [41]. The indoor temperature evolution is described by

$$C_i \left(Tem_{i,t+1}^{\text{in}} - Tem_{i,t}^{\text{in}} \right) = \frac{A_i}{R_i} \left(Tem_{i,t}^{\text{out}} - Tem_{i,t}^{\text{in}} \right) + \eta_i^{\text{h}} P_{i,t}^{\text{svr}} - \eta_i^{\text{c}} P_{i,t}^{\text{col}}, \quad \forall i \in DC, \ \forall t \in T \tag{11}$$

where $Tem_{i,t}^{\text{in}}$ and $Tem_{i,t}^{\text{out}}$ denote the indoor and outdoor temperatures of DC i at time t. A_i, R_i, and C_i denote the effective heat-transfer area, the area-normalised thermal resistance, and the effective thermal capacitance of DC i, respectively. The ratio A_i/R_i therefore represents the equivalent thermal conductance between the indoor and outdoor environments. The coefficient η_i^{h} denotes the fraction of server electrical power converted into heat, while η_i^{c} denotes the coefficient of performance (COP) of the cooling system, defined as the amount of heat removed per unit of electrical power consumed by the cooling equipment. Since almost all electrical power consumed by the servers is ultimately dissipated as heat inside the DC, the term $\eta_i^{\text{h}} P_{i,t}^{\text{svr}}$ represents the server heat generation, while $\eta_i^{\text{c}} P_{i,t}^{\text{col}}$ represents the cooling effect provided by the cooling system. The indoor temperature must remain within the admissible range:

$$\underline{Tem}_i^{\text{in}} \le Tem_{i,t}^{\text{in}} \le \overline{Tem}_i^{\text{in}}, \forall i \in DC, \forall t \in T \tag{12}$$

It is worth noting that the first-order thermal model is adopted because this study focuses on system-level restoration scheduling over hourly time intervals (similar to [16,40]), rather than short-term airflow dynamics or rack-level temperature distributions. At this temporal and spatial resolution, the model retains the dominant thermal inertia and the main coupling among ambient temperature, server heat generation, and cooling power. A higher-order multi-zone or airflow-based model would require detailed facility-specific parameters and introduce additional states and computational complexity into the two-stage robust optimisation problem, while providing limited additional information for the hourly restoration decisions considered here. Therefore, the adopted model provides an appropriate balance between thermal representation and computational tractability. Nevertheless, detailed rack-level thermal dynamics may be incorporated in future studies involving faster operational timescales.

Therefore, the total electrical demand of DCMG i is given by:

$$P_{i,t}^{\text{dc}} = P_{i,t}^{\text{svr}} + P_{i,t}^{\text{col}}, \forall i \in DC, \forall t \in T \tag{13}$$

The cooling power is not prescribed as a fixed ratio of the server power. Instead, $P_{i,t}^{\text{col}}$ is a continuous recourse variable jointly determined by the thermal balance (11), the cooling-capacity constraint (10), and the indoor-temperature limits (12). The heat-removal rate provided by the cooling system is

$$Q_{i,t}^{\text{cool}} = \eta_i^{\text{c}} P_{i,t}^{\text{col}}, \forall i \in DC, \forall t \in T \tag{14}$$

Therefore, Eq. (11) can equivalently be rearranged as

$$P_{i,t}^{\text{col}} = \frac{1}{\eta_i^{\text{c}}} \left[\frac{A_i}{R_i} \left(Tem_{i,t}^{\text{out}} - Tem_{i,t}^{\text{in}} \right) + \eta_i^{\text{h}} P_{i,t}^{\text{svr}} - C_i \left(Tem_{i,t+1}^{\text{in}} - Tem_{i,t}^{\text{in}} \right) \right] \tag{15}$$

where the required cooling electricity depends on the server heat generation, heat exchange with the outdoor environment, and the scheduled indoor-temperature trajectory.

3.1.6. Energy storage system

To ensure a degree of reliability for the power supply, DCMGs can be equipped with battery energy storage systems (ESSs), which guarantee an uninterrupted power supply and provide support for the load restoration of the whole distribution network after extreme events. Therefore, an ESS can be configured for each DCMG, where the detailed mathematical formulations are found in Wang et al.,Abdolahi et al. [42,43].

3.2. MPS routing and scheduling model

MEGs and MESSs are considered as two types of MPSs that can choose to connect to candidate nodes, e.g., MESS stations (MSs) [29]. Following [29,44], this paper assumes that both MESSs and MEGs have black-start capability during the load restoration process. The role difference between MESSs and MEGs is that MESSs can charge at one location with sufficient power supply and then discharge power at another location suffering load shedding, while MEGs can only provide power supply to the power network. In other words, MESSs play a role similar to demand-side response, whereas MEGs play a role similar to traditional generators but with mobility features.

3.2.1. Routing model

The routing behaviours of mobile units $i \in \mathcal{MPS}$ including both MEGs and MESSs can be formulated in the same manner [28,30], and are restricted by

$$\sum_{n \in N} u_{i,n,t} \le 1, \forall i \in \mathcal{MPS}, \forall t \in T \tag{16}$$

$$\sum_{\tau=t}^{\min\left(t+T_{mn,t}^{rd}, T\right)} u_{i,m,\tau} \le (1 - u_{i,n,t}) \cdot \min\left(T_{mn,t}^{rd}, T - t\right), \forall i \in \mathcal{MPS}, \forall n \in N, \forall m \in N \backslash \{i\}, \forall t \in T \tag{17}$$

where constraint (16) shows that a mobile unit i can be connected to only one transport node at each time step, and the binary variable $u_{i,n,t} \in \{0,1\}$ indicates whether the mobile unit i is connected to node n ($u_{i,n,t} = 1$) or not ($u_{i,n,t} = 0$) at time t. Constraint (17) ensures reasonable routing of the mobile unit i between different transport nodes, where $T_{mn,t}^{rd}$ refers to the time period required to route from node n to node m at time t.

3.2.2. Scheduling model

When connected to the power network, MPSs make scheduling decisions (i.e., MEG g power output $P_{g,n,t}^{eg}$, MESS k charging power $P_{k,n,t}^{esc}$ and discharging power $P_{k,n,t}^{esd}$) in MS n. The operation models of MPSs can be found below:

1) MEGs are constrained by their active and reactive power output limits

$$\underline{P}_g^{eg} \le P_{g,n,t}^{eg} \le \overline{P}_g^{eg}, \forall g \in \mathcal{MEG}, \forall n \in N_{ms}, \forall t \in T \tag{18}$$

$$\underline{Q}_g^{eg} \le Q_{g,n,t}^{eg} \le \overline{Q}_g^{eg}, \forall g \in \mathcal{MEG}, \forall n \in N_{ms}, \forall t \in T \tag{19}$$

where $P_{g,n,t}^{eg}$ and $Q_{g,n,t}^{eg}$ correspond to the active and reactive power outputs of MEG g in MS n at time step t, respectively.

2) The power operation model of MESSs can be appropriately presented via

$$0 \le P_{k,n,t}^{esc} \le u_{k,t}^{es} \overline{P}_k^{es}, \forall k \in \mathcal{MES}, \forall n \in N_{ms}, \forall t \in T \tag{20}$$

$$-\overline{P}_k^{es} \left(1 - u_{k,t}^{es}\right) \le P_{k,n,t}^{esd} \le 0, \forall k \in \mathcal{MES}, \forall n \in N_{ms}, \forall t \in T \tag{21}$$

$$\underline{S}_k^{es} \le S_{k,t}^{es} \le \overline{S}_k^{es}, \forall k \in \mathcal{MES}, \forall t \in T \tag{22}$$

$$S_{k,t+1}^{es} = S_{k,t}^{es} + \frac{P_{k,n,t}^{esc} \eta_k^{esc} + P_{k,n,t}^{esd} / \eta_k^{esd}}{\overline{E}_k^{es}}, \forall k \in \mathcal{MES}, \forall n \in N_{ms}, \forall t \in T \tag{23}$$

where constraints (20) and (21) restrict the maximum charging and discharging power of MESS k in MS n at time step t. The binary variable $u_{k,t}^{es} \in \{0,1\}$ introduced in constraints (20) and (21) indicates the charging ($u_{k,t}^{es} = 1$) or discharging ($u_{k,t}^{es} = 0$) behavior of MESS k at time step t. It is noted that these two behaviours cannot happen simultaneously. Constraint (22) limits the minimum and maximum battery state-of-charge (SoC) level of MESS k, while its dynamic transition between two consecutive time steps is presented in (23), given the charging/discharging powers $P_{k,t}^{esc}$, $P_{k,t}^{esd}$ and efficiencies $\eta_k^{esc}, \eta_k^{esd}$.

3.3. Power network model

The power network operation is fully modelled by a linearised OPF algorithm with the objective of the maximisation of load restoration

towards resilience enhancement, which can be illustrated below:

$$\sum_{g\in\mathcal{B}_{dg}} P^{dg}_{g,t} + \sum_{g\in\mathcal{B}_{eg}} P^{eg}_{g,t} + \sum_{w\in\mathcal{B}_{pv}} P^{pv}_{w,t} = \sum_{d\in\mathcal{B}_{ed}} P^{ed}_{d,t} + \sum_{k\in\mathcal{B}_{es}} P^{es}_{k,t} + \sum_{i\in\mathcal{B}_{dc}} P^{dc}_{i,t} - \sum_{(p,b)\in L} P_{pb,t} + \sum_{(b,p)\in L} P_{bp,t},\ \forall b\in\mathcal{B},\forall t\in T \tag{24}$$

$$\sum_{g\in\mathcal{B}_{dg}} Q^{dg}_{g,t} + \sum_{g\in\mathcal{B}_{eg}} Q^{eg}_{g,t} + \sum_{w\in\mathcal{B}_{pv}} Q^{pv}_{w,t} = \sum_{d\in\mathcal{B}_{ed}} Q^{ed}_{d,t} - \sum_{(p,b)\in L} Q_{pb,t} + \sum_{(b,p)\in L} Q_{bp,t},\ \forall b\in\mathcal{B},\forall t\in T \tag{25}$$

$$\underline{P}^{dg}_{g} \le P^{dg}_{g,t} \le \overline{P}^{dg}_{g}, \forall g\in\mathcal{DG},\forall t\in T \tag{26}$$

$$\underline{Q}^{dg}_{g} \le Q^{dg}_{g,t} \le \overline{Q}^{dg}_{g}, \forall g\in\mathcal{DG},\forall t\in T \tag{27}$$

$$0 \le P^{pv}_{w,t} \le \tilde{P}^{pv}_{w,t}, \forall w\in\mathcal{PV}^{bs},\forall t\in T \tag{28}$$

$$\left(P^{pv}_{w,t}\right)^2 + \left(Q^{pv}_{w,t}\right)^2 \le \left(\overline{S}^{pv}_{w}\right)^2, \forall w\in\mathcal{PV}^{bs},\forall t\in T \tag{29}$$

$$0 \le P^{pv}_{w,t} \le e_{b,t}\tilde{P}^{pv}_{w,t},\ \forall b\in\mathcal{B},\ \forall g\in\mathcal{B}_{pv}\cap\mathcal{PV}^{nbs},\forall t\in T \tag{30}$$

$$\left(P^{pv}_{w,t}\right)^2 + \left(Q^{pv}_{w,t}\right)^2 \le e_{b,t}\left(\overline{S}^{pv}_{w}\right)^2, \forall b\in\mathcal{B},\forall w\in\mathcal{B}_{pv}\cap\mathcal{PV}^{nbs},\forall t\in T \tag{31}$$

$$P^{ed}_{d,t} \le e_{b,t}\overline{P}^{ed}_{d,t},\ \forall b\in\mathcal{B},\forall d\in\mathcal{B}_{ed},\forall t\in T \tag{32}$$

$$Q^{ed}_{d,t} \le e_{b,t}\overline{Q}^{ed}_{d,t},\ \forall b\in\mathcal{B},\forall d\in\mathcal{B}_{ed},\forall t\in T \tag{33}$$

$$e_{b,t}\underline{V}^2 \le V^2_{b,t} \le e_{b,t}\overline{V}^2,\ \forall b\in\mathcal{B},\forall t\in T \tag{34}$$

$$P^2_{bp,t} + Q^2_{bp,t} \le y_{bp,t}\cdot\overline{S}^2_{bp},\ \forall(b,p)\in\mathcal{L},\forall t\in T \tag{35}$$

$$V^2_{b,t} - V^2_{p,t} \le 2\cdot(r_{bp}P_{bp,t} + x_{bp}Q_{bp,t}) + (1-y_{bp,t})\cdot M,\ \forall(b,p)\in\mathcal{L},\forall t\in T \tag{36}$$

$$V^2_{b,t} - V^2_{p,t} \ge 2\cdot(r_{bp}P_{bp,t} + x_{bp}Q_{bp,t}) + (y_{bp,t}-1)\cdot M,\ \forall(b,p)\in\mathcal{L},\forall t\in T \tag{37}$$

$$\sum_{(b,p)\in L} y_{bp,t} = |B| - \left(\sum_{b\in\mathcal{B}}(1-e_{b,t}) + N_{bs}\right), \forall t\in T \tag{38}$$

$$\sum_{a\in\mathcal{B}_{bs}} F^{s}_{a,t} + \sum_{(p,b)\in L} F_{pb,t} - \sum_{(b,p)\in L} F_{bp,t} = e_{b,t},\ \forall b\in\mathcal{B},\forall t\in T \tag{39}$$

$$-y_{bp,t}\overline{F}_{bp} \le F_{bp,t} \le y_{bp,t}\overline{F}_{bp}, \forall(b,p)\in\mathcal{L},\forall t\in T \tag{40}$$

where the OPF constraints are formulated using the LinDistFlow model [45]. The active and reactive power balances at bus b are given in (24)-(25), where $\mathcal{B}_{eg}$, $\mathcal{B}_{es}$, $\mathcal{B}_{ed}$, $\mathcal{B}_{dg}$, and $\mathcal{B}_{pv}$ correspond to the sets of MEG g, MESS k, restored load d, diesel generators (DGs) g, and photovoltaics (PVs) w located at bus b, respectively. The active and reactive power outputs of DG g are constrained by (26) and (27), respectively, while the output limits of grid-forming and grid-following PVs are presented in (28)-(29) and (30)-(31), respectively [44]. Note that the set of PV units is divided into the grid-forming and grid-following subsets, i.e., $\mathcal{PV} = \mathcal{PV}^{bs}\cup\mathcal{PV}^{nbs}$ and $\mathcal{PV}^{bs}\cap\mathcal{PV}^{nbs} = \varnothing$. Grid-forming PV units are assumed to have black-start capability and can establish the voltage and frequency reference for an islanded section, as indicated in [44]. In contrast, grid-following PV units require an existing voltage and frequency reference and can operate only when their connected buses have been energised. Accordingly, the two types of PV units are subject to different restoration constraints. In detail, constraints (28) and (29) define the available active-power and apparent-power limits of grid-forming PV units. Since these units have black-start capability, their outputs are not conditional on a previously energised bus. The outputs of grid-following PV units in (30) and (31) are linked to the bus energisation variable $e_{b,t}$. Therefore, a grid-following PV unit can inject active and reactive power only after its connected bus has been energised by a grid-forming or other black-start resource. The status of load d is restricted by constraints (32) and (33), where the binary variable $e_{b,t}$ corresponds to the energised status of bus b (1 if energised, 0 otherwise). The voltage and power flow limits are shown in (34) and (35), respectively, while the linearised power flow constraints are expressed in (36)-(37). The binary variable $y_{bp,t}$ in (35)-(37) indicates the energised status of line (b,p) (1 if energised, 0 otherwise), which is determined by the reconfiguration switch, while M refers to a large positive number used to relax constraints (36)-(37) for disconnected or damaged lines.

To coordinate with MPS dispatch, the distribution network can be dynamically reconfigured through smart switch operations. The resulting topology must satisfy radiality, enforced by the virtual network constraints (38)-(40) [44,46]. According to [47], radiality requires that: i) each island is connected; and ii) the number of energised lines equals the number of buses minus the number of islands. Furthermore, an additional condition is introduced in [46] to ensure that de-energised or cut-off buses are not connected to each other due to the lack of black-start resources. Specifically, constraint (38) maintains the network radiality during the network reconfiguration process, which shows that the number of lines is equal to the number of buses minus the number of possible islands containing black-start resources and de-energised buses [46]. Constraint (39) gives the virtual-network nodal balance, where bus b is energised if its virtual load is served, i.e., $e_{b,t} = 1$. Here, $\mathcal{B}_{bs}$ is the set of power sources with black-start capabilities, such as buses connected to DGs, grid-forming PVs, and MPSs, located at bus b. Constraint (40) links real and virtual networks through the line connection status $y_{bp,t}$.

3.4. Objective function

The objective function of the proposed resilience-oriented coordination problem is to minimise the load shedding costs of the whole distribution network, which is illustrated below:

$$\min_{\Xi^{dc},\Xi^{mps},\Xi^{net}} \sum_{d\in D}\sum_{t\in T} c^{ls}_{d} P^{ls}_{d,t} \tag{41}$$

where $P^{ls}_{d,t}$ is the load shedding quantity of load d at time t. Note that the power system captures both essential and non-essential loads to highlight the primary objective of load restoration [42], and c^{ls}_{d} corresponds to the load shedding cost of load d in the distribution network. Ξ^{dc}, Ξ^{mps}, Ξ^{net} represent the decision variables related to DCMGs, MPSs, and the distribution network, respectively.

Battery degradation is not explicitly included in the present formulation. Although battery ageing is important for long-term asset management and routine economic operation, the present study focuses on short-term emergency restoration over a 24-hour horizon following an extreme event. Under such conditions, restoring interrupted loads is prioritised over the comparatively small degradation cost associated with a limited number of emergency charging and discharging operations [30]. Nevertheless, the MESS power limits, state-of-charge bounds, charging/discharging efficiencies, and mutually exclusive operating states are retained to ensure physically feasible battery operation.

4. Asymmetric data-driven approach for uncertainty modelling

The uncertainty set in this paper is constructed from historical forecast errors of PV generation, demand, and ambient temperature. Different from an original symmetric channel-wise treatment suggested in the existing literature such as [39,40], the proposed approach adopts an asymmetric uncertainty description so that upward and downward deviations are modelled separately. The overall procedure of the proposed uncertainty modelling approach is illustrated in Fig. 2.

4.1. Data-driven calibration of uncertainty bounds

Before constructing the uncertainty set, historical forecast-error samples are used to calibrate the robust deviation bounds [48]. Following

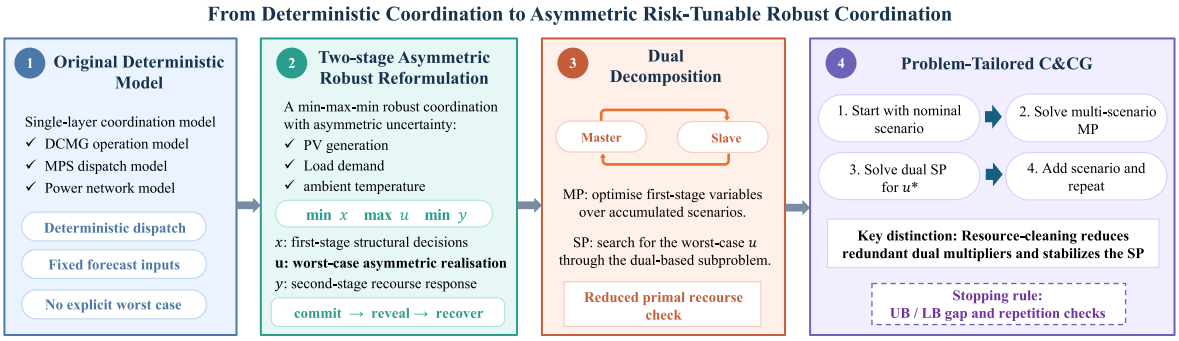


Fig. 2. Illustration of the transition from the deterministic coordination model to the proposed two-stage asymmetric risk-tunable robust formulation.

the data-driven risk-tunable method in [40], the empirical distribution of the standardised forecast errors is enclosed by a Wasserstein ambiguity set:

$$\mathcal{P}^{\kappa} = \left\{ \mathbb{P} : W\left(\mathbb{P}, \hat{\mathbb{P}}^{\kappa}\right) \le \varepsilon^{\kappa} \right\} \tag{42}$$

where $\hat{\mathbb{P}}^{\kappa}$ is the empirical distribution, $W(\cdot)$ denotes the Wasserstein distance, and ε^{κ} is determined by the confidence level ϕ and the number of samples.

The risk-tunable bound is obtained by limiting the worst-case probability that the standardised forecast error exceeds the uncertainty boundary:

$$\sup_{\mathbb{P}\in\mathcal{P}^{\kappa}} \mathbb{P}\left[|\xi^{\kappa}| > \bar{\xi}^{\kappa}\right] \le 1-\beta \tag{43}$$

where β is the risk level. The resulting bound $\bar{\xi}^{\kappa}$ is then mapped back to the original physical scale to obtain the robust deviation Δ^{κ}. In this work, this calibration is applied separately to positive and negative forecast error samples, leading to asymmetric bounds $\Delta^{\kappa,+}$ and $\Delta^{\kappa,-}$.

4.2. Risk-tunable uncertainty set

Based on the calibrated asymmetric bounds, let $\kappa \in \{\mathrm{pv}, \mathrm{ld}, \mathrm{tp}\}$ denote the uncertainty channel corresponding to PV generation, demand, and ambient temperature, respectively. For channel κ, let $\mathcal{Z}^{\kappa} = \{\zeta_n^{\kappa}\}_{n=1}^{N_{\kappa}}$ be the set of historical relative forecast error samples. For PV generation and demand, the relative error is calculated as $\zeta_n^{\kappa} = (x_n^{\kappa} - \hat{x}_n^{\kappa})/\hat{x}_n^{\kappa}$, where samples with zero nominal values are excluded. For ambient temperature, the relative error is calculated on the Kelvin scale as $\zeta_n^{\mathrm{tp}} = (T_n^{\mathrm{out}} - \hat{T}_n^{\mathrm{out}})/(\hat{T}_n^{\mathrm{out}} + 273.15)$.

The positive and negative error subsets are extracted as

$$\mathcal{Z}^{\kappa,+} = \left\{ \zeta_n^{\kappa} \in \mathcal{Z}^{\kappa} : \zeta_n^{\kappa} > 0 \right\}, \tag{44a}$$

$$\mathcal{Z}^{\kappa,-} = \left\{ -\zeta_n^{\kappa} : \zeta_n^{\kappa} \in \mathcal{Z}^{\kappa},\ \zeta_n^{\kappa} < 0 \right\}. \tag{44b}$$

For each subset, the empirical mean and standard deviation are denoted by $\mu^{\kappa,+}$, $\sigma^{\kappa,+}$, $\mu^{\kappa,-}$, and $\sigma^{\kappa,-}$, respectively. Therefore, the corresponding standardised samples are

$$\xi_n^{\kappa,+} = \frac{\zeta_n^{\kappa,+} - \mu^{\kappa,+}}{\sigma^{\kappa,+}} \tag{45a}$$

$$\xi_n^{\kappa,-} = \frac{\zeta_n^{\kappa,-} - \mu^{\kappa,-}}{\sigma^{\kappa,-}} \tag{45b}$$

Accordingly, the positive and negative subsets are separately calibrated to obtain the following asymmetric robust deviations:

$$\Delta^{\kappa,+} = \mu^{\kappa,+} + \sigma^{\kappa,+}\bar{\xi}^{\kappa,+} \tag{46a}$$

$$\Delta^{\kappa,-} = \mu^{\kappa,-} + \sigma^{\kappa,-}\bar{\xi}^{\kappa,-} \tag{46b}$$

where $\bar{\xi}^{\kappa,+}$ and $\bar{\xi}^{\kappa,-}$ are the risk-tunable bounds recovered from the positive and negative sample sets, respectively. Since the calibration is based on relative forecast errors, $\Delta^{\kappa,+}$ and $\Delta^{\kappa,-}$ are dimensionless relative deviation bounds.

Accordingly, the uncertain realisations are represented in relative deviation form as

$$\tilde{P}_{w,t}^{\mathrm{pv}} = \hat{P}_{w,t}^{\mathrm{pv}}\left(1 + \Delta^{\mathrm{pv},+}u_{w,t}^{\mathrm{pv},+} - \Delta^{\mathrm{pv},-}u_{w,t}^{\mathrm{pv},-}\right), \forall w \in \mathcal{PV},\ \forall t \in \mathcal{T} \tag{47a}$$

$$\tilde{P}_{d,t}^{\mathrm{ed}} = \hat{P}_{d,t}^{\mathrm{ed}}\left(1 + \Delta^{\mathrm{ld},+}u_{d,t}^{\mathrm{ld},+} - \Delta^{\mathrm{ld},-}u_{d,t}^{\mathrm{ld},-}\right), \forall d \in \mathcal{D},\ \forall t \in \mathcal{T} \tag{47b}$$

$$\tilde{T}_{i,t}^{\mathrm{out}} = \hat{T}_{i,t}^{\mathrm{out}} + \left(\hat{T}_{i,t}^{\mathrm{out}} + 273.15\right)\left(\Delta^{\mathrm{tp},+}u_{i,t}^{\mathrm{tp},+} - \Delta^{\mathrm{tp},-}u_{i,t}^{\mathrm{tp},-}\right), \forall i \in \mathcal{DC},\ \forall t \in \mathcal{T} \tag{47c}$$

where $\hat{P}_{w,t}^{\mathrm{pv}}$, $\hat{P}_{d,t}^{\mathrm{ed}}$, and $\hat{T}_{i,t}^{\mathrm{out}}$ are the nominal forecasts, and $u_{\cdot,t}^{(\cdot),+}, u_{\cdot,t}^{(\cdot),-} \in \{0,1\}$ are binary direction variables. Therefore, the corresponding physical deviation is proportional to the nominal value of each unit rather than being imposed as a common absolute deviation.

The uncertainty direction at each location and time step t is restricted by:

$$u_{w,t}^{\mathrm{pv},+} + u_{w,t}^{\mathrm{pv},-} \le 1, \forall w \in \mathcal{PV},\ \forall t \in \mathcal{T} \tag{48a}$$

$$u_{d,t}^{\mathrm{ld},+} + u_{d,t}^{\mathrm{ld},-} \le 1, \forall d \in \mathcal{D},\ \forall t \in \mathcal{T} \tag{48b}$$

$$u_{i,t}^{\mathrm{tp},+} + u_{i,t}^{\mathrm{tp},-} \le 1, \forall i \in \mathcal{DC},\ \forall t \in \mathcal{T} \tag{48c}$$

To balance robustness and conservativeness, the channel-wise cardinality budgets are imposed to limit the number of time intervals in which each uncertain parameter can deviate from its nominal value:

$$\sum_{t\in\mathcal{T}}\left(u_{w,t}^{\mathrm{pv},+} + u_{w,t}^{\mathrm{pv},-}\right) \le \Gamma^{\mathrm{pv}}, \forall w \in \mathcal{PV} \tag{49a}$$

$$\sum_{t\in\mathcal{T}}\left(u_{d,t}^{\mathrm{ld},+} + u_{d,t}^{\mathrm{ld},-}\right) \le \Gamma^{\mathrm{ld}}, \forall d \in \mathcal{D} \tag{49b}$$

$$\sum_{t\in\mathcal{T}}\left(u_{i,t}^{\mathrm{tp},+} + u_{i,t}^{\mathrm{tp},-}\right) \le \Gamma^{\mathrm{tp}}, \forall i \in \mathcal{DC} \tag{49c}$$

Compared with existing works (e.g., [40]), the data-driven risk-tunable uncertainty modelling in this work is extended to an asymmetric setting. This modification is physically meaningful for resilience-oriented scheduling, since PV downward deviation is generally more harmful than PV upward deviation, while load increase and temperature rise are typically more critical than their opposite directions. Moreover, the relative-deviation representation accounts for the different capacities and operating levels of individual units.

4.3. Two-stage robust reformulation

Based on the above risk-tunable uncertainty set, the resilience-oriented coordination problem proposed in Section 3 can be formulated as a two-stage robust optimisation model, where $\mathcal{U}$ is used to represent the asymmetric data-driven risk-tunable uncertainty set defined in

(47a)-(49a),(49b) and (49c). The overall problem is written as

$$\min_{\mathbf{x}\in\mathcal{X}} \max_{\mathbf{u}\in\mathcal{U}} \min_{\mathbf{y}\in\mathcal{Y}(\mathbf{x},\mathbf{u})} f(\mathbf{y},\mathbf{u}) \tag{50}$$

where $\mathbf{u}$ denotes the uncertainty realisation. $\mathbf{x}$ denotes the first-stage binary decisions, including MPS routing, network reconfiguration, and load energisation, while $\mathbf{y}$ denotes the second-stage continuous recourse decisions, including generation dispatch, power flow, MESS charging/discharging power, served load, DC workload transfer, server dispatch, cooling power, temperature states, etc.

Following the objective defined in (41), the recourse objective can be defined to penalise the unserved conventional load:

$$f(\mathbf{y},\mathbf{u}) = \sum_{d\in\mathcal{D}} \sum_{t\in\mathcal{T}} c_d^{\mathrm{ls}} P_{d,t}^{ls} \tag{51}$$

4.4. Problem-tailored C&CG approach

To solve the two-stage robust optimisation problem in (50), a C&CG algorithm [7] is adopted. At iteration k, the master problem retains all previously identified worst-case scenarios and their corresponding recourse variables and constraints. Consequently, the first-stage decisions are evaluated against the accumulated set of adverse scenarios rather than only the most recently generated realisation. After the master problem is solved, the first-stage solution is fixed, and the slave problem identifies a new worst-case uncertainty realisation. If the convergence criterion is not satisfied, the new scenario and its associated recourse block are added to the master problem for the next iteration. For the integrated resilience coordination model considered in this work, a problem-tailored recourse reformulation is introduced before applying the standard C&CG procedure. After fixing the first-stage routing and network-topology decisions, redundant topology-construction constraints and auxiliary variables are removed, while the relevant quadratic constraints are polygonally linearised. The resulting second-stage problem is reduced to an LP that permits valid linear dualisation. Specifically, the problem is decomposed into the following master and slave problems:

4.4.1. Master problem

At iteration k, let Ω^k be the set of accumulated worst-case scenarios. Accordingly, the master problem simultaneously considers all scenarios in Ω^k and can be written as:

$$\min_{\mathbf{x},\eta,\{\mathbf{y}^\omega\}_{\omega\in\Omega^k}} \eta \tag{52}$$

subject to

$$\eta \geq f(\mathbf{y}^\omega,\omega), \forall\omega\in\Omega^k \tag{53}$$

$$\mathbf{y}^\omega \in \mathcal{Y}(\mathbf{x},\omega), \forall\omega\in\Omega^k \tag{54}$$

where the common first-stage vector $\mathbf{x}$ is shared across all accumulated scenarios, while each scenario ω corresponds to its own recourse variables $\mathbf{y}^\omega$.

4.4.2. Topology-cleaned recourse LP

After solving the master problem, the first-stage binary solution $\mathbf{x}^*$ is fixed. In the present model, $\mathbf{x}^*$ includes the MPS routing decisions $u_{i,n,t}$, line-connection decisions $y_{bp,t}$, bus-energisation decisions $e_{b,t}$, and other binary structural decisions. For a given uncertainty realisation $\mathbf{u}$, the original second-stage recourse problem is defined as

$$Q(\mathbf{x}^*,\mathbf{u}) = \min_{\mathbf{y}\in\mathcal{Y}(\mathbf{x}^*,\mathbf{u})} f(\mathbf{y},\mathbf{u}) \tag{55}$$

To obtain an LP recourse formulation, the quadratic constraints (29), (31), and (35) are first approximated by N-segment polygonal constraints [49], as detailed in Appendix A. Let $\mathcal{Y}_N^{\mathrm{F}}(\mathbf{x}^*,\mathbf{u})$ denote the resulting full polygonally linearised recourse set. The subscript N is introduced to distinguish this polygonally linearised model from the original recourse model containing quadratic constraints.

The variables in the full polygonally linearised recourse problem are partitioned into continuous operational variables $\mathbf{z}$ and topology-certification variables $\mathbf{v}$. The vector $\mathbf{z}$ contains the continuous variables associated with power dispatch, power flow, load restoration, MPS charging and discharging, DC workload allocation, server operation, cooling power, and indoor temperature. The vector $\mathbf{v}$ contains only the virtual black-start injections $F_{a,t}^s$ and virtual line flows $F_{bp,t}$ introduced in (39) and (40) to certify the radiality of the first-stage network topology. Accordingly, the full polygonally linearised recourse problem can be expressed as

$$\begin{aligned} Q_N^{\mathrm{F}}(\mathbf{x}^*,\mathbf{u}) = \min_{\mathbf{z},\mathbf{v}} \quad & f(\mathbf{z},\mathbf{u}) \\ \text{s.t.} \quad & \mathbf{z} \in \mathcal{Y}_N^{\mathrm{op}}(\mathbf{x}^*,\mathbf{u}) \\ & \mathbf{v} \in \mathcal{Y}^{\mathrm{top}}(\mathbf{x}^*) \end{aligned} \tag{56}$$

where $\mathcal{Y}_N^{\mathrm{op}}(\mathbf{x}^*,\mathbf{u})$ contains all physical and operational constraints under the fixed first-stage decisions, while $\mathcal{Y}^{\mathrm{top}}(\mathbf{x}^*)$ contains only the first-stage structural and topology-certification constraints.

More specifically, after $\mathbf{x}^*$ is fixed, the following components are removed from the recourse optimisation: i) the routing-feasibility constraints (16) and (17), since these constraints contain only the fixed MPS routing variables $u_{i,n,t}$; ii) the radiality-count constraint (38), since the line-connection decisions $y_{bp,t}$ and bus-energisation decisions $e_{b,t}$ have already been determined and validated in the master problem; and iii) the virtual black-start injections $F_{a,t}^s$, the virtual line flows $F_{bp,t}$, and their associated topology-certification constraints (39) and (40).

The fixed variables $u_{i,n,t}$, $y_{bp,t}$, and $e_{b,t}$ are not discarded from the operational model. Their optimised values are passed to the recourse problem as fixed parameters wherever they appear in the physical operating constraints. In particular, the nodal active and reactive power balances (24)–(25), the DG operating constraints (26) and (27), the PV operating constraints (28)–(31), the load-energisation constraints (32) and (33), the voltage constraints (34), the line-flow limits (35), the voltage-drop constraints (36) and (37), and all DCMG operational and thermal constraints remain in the recourse problem.

The removal of the topology-certification block is exact for every $\mathbf{u}\in\mathcal{U}$. First, $\mathcal{Y}^{\mathrm{top}}(\mathbf{x}^*)$ depends only on the fixed first-stage solution and is independent of both the operational variables $\mathbf{z}$ and the uncertainty realisation $\mathbf{u}$. Second, because $\mathbf{x}^*$ is a feasible solution obtained from the master problem, the topology-certification set is nonempty, i.e., $\mathcal{Y}^{\mathrm{top}}(\mathbf{x}^*)\neq\emptyset$. Third, the topology-certification variables $\mathbf{v}$ do not appear in the recourse objective. Therefore, minimisation over $\mathbf{v}$ does not affect either the feasible set of $\mathbf{z}$ or the recourse objective value. The topology-cleaned recourse problem can consequently be written as

$$Q_N^{\mathrm{R}}(\mathbf{x}^*,\mathbf{u}) = \min_{\mathbf{z}\in\mathcal{Y}_N^{\mathrm{op}}(\mathbf{x}^*,\mathbf{u})} f(\mathbf{z},\mathbf{u}) \tag{57}$$

and satisfies

$$Q_N^{\mathrm{F}}(\mathbf{x}^*,\mathbf{u}) = Q_N^{\mathrm{R}}(\mathbf{x}^*,\mathbf{u}), \forall\mathbf{u}\in\mathcal{U} \tag{58}$$

Thus, no variable or constraint is removed according to whether it is inactive or weakly binding at the current solution. The removed components form a structurally separable topology-certification block that is independent of the uncertainty realisation and has already been enforced in the master problem. All uncertainty-dependent and physical operating constraints are retained in the recourse problem.

It should also be clarified that the equivalence in (58) concerns only the removal of the topology-certification block under the same N-segment polygonal formulation. The polygonal representation itself remains an approximation of the original quadratic constraints. The

resulting topology-cleaned LP contains fewer auxiliary variables, constraints, and dual multipliers, thereby facilitating the construction and solution of the dual slave problem.

4.4.3. Dual slave problem

The reduced LP recourse problem suggested in (57) can be dualised to form a slave problem, which is expressed as

$$\max_{\mathbf{u}\in\mathcal{U},\,\lambda\in\Lambda(\mathbf{x}^*)} g(\mathbf{x}^*,\mathbf{u},\lambda) \tag{59}$$

where $\Lambda(\mathbf{x}^*)$ represents the dual feasible region and $g(\cdot)$ is the dual objective. λ corresponds to the dual multipliers associated with the uncertainty-coupled constraints. Note that the uncertainty is represented by binary variables and enters the PV limits, load limits, and thermal balance constraints. Since the resulting recourse problem is an LP, its dual feasible region is polyhedral. Therefore, the bilinear terms between the binary variables and dual variables in the slave problem can be linearised using the Big-M method [7].

In the present implementation, the asymmetric uncertainty set leads to different coefficients for upward and downward perturbations. For example, the PV channel uses $\Delta^{\mathrm{pv},+}$ and $\Delta^{\mathrm{pv},-}$ separately, while the load and temperature channels use $\Delta^{\mathrm{ld},+}$, $\Delta^{\mathrm{ld},-}$ and $\Delta^{\mathrm{tp},+}$, $\Delta^{\mathrm{tp},-}$, respectively. Therefore, compared with the conventional symmetric formulation in [40], the proposed slave problem explicitly distinguishes the worst-case impacts of positive and negative deviations.

4.4.4. Iterative procedure

The overall procedure of the proposed iterative C&CG algorithm is summarised below:

Algorithm 1 Problem-Tailored C&CG Algorithm.

1: Initialise the accumulated scenario set Ω^0 with the nominal forecast scenario
2: Set iteration counter $k \leftarrow 0$
3: **repeat**
4: Solve the multi-scenario master problem (52)–(54) and obtain the first-stage decision $\mathbf{x}^*$
5: Fix $\mathbf{x}^*$ and solve the dual slave problem (59) to identify the worst-case uncertainty realisation $\mathbf{u}^*$
6: Solve the primal reduced recourse LP under $(\mathbf{x}^*,\mathbf{u}^*)$ and evaluate the corresponding worst-case cost
7: Update the upper and lower bounds UB and LB
8: Update the scenario set $\Omega^{k+1} = \Omega^k \cup \{\mathbf{u}^*\}$
9: Set $k \leftarrow k+1$
10: **until** $\frac{UB - LB}{\max\{1, |UB|\}} \le \varepsilon$ or the same first-stage solution or worst-case scenario reappears

5. Input data and experimental setup

5.1. Network structure and input data

Case studies are conducted on a modified IEEE 33-bus distribution network with 3 DCMGs, 1 MESS, and 1 MEG, where the power network structure and the detailed road network including 5 MS stations are illustrated in Fig. 3(a) and (b), respectively. Conventional DGs and RESs are appropriately allocated in the distribution network. The RES and demand data are collected from Ausgrid [50], which are illustrated in Fig. 4. The technical parameters related to the investigated power network can be found in [30], while the operation data for conventional DGs are collected from [42]. The event is assumed to occur at $t = 0$, and the resulting line outages persist throughout the 24-hour restoration horizon, following the same practice in [28,30]. The set of tripped lines remains fixed, while the remaining in-service switchable lines can be reconfigured over time.

5.2. Technical parameters

The main technical parameters of the DCs in DCMGs and MPSs used in the modified IEEE 33-bus distribution network are summarised in Tables 1–3. The curtailment costs for essential and non-essential loads are set to 2.5 £/(kW·h) and 1.5 £/(kW·h), respectively. The workload profiles and ambient temperature profiles of three DCs are illustrated in Fig. 5, while the uncertainty budgets for renewable generation, demand, and ambient temperature are all set to 6 time periods.

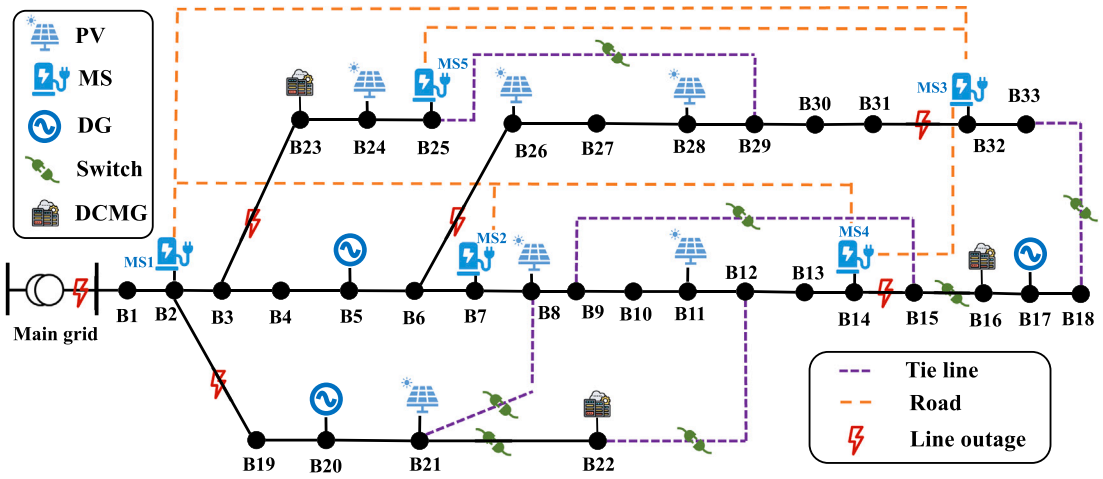
6. Case studies

6.1. Analysis of the coordination performance

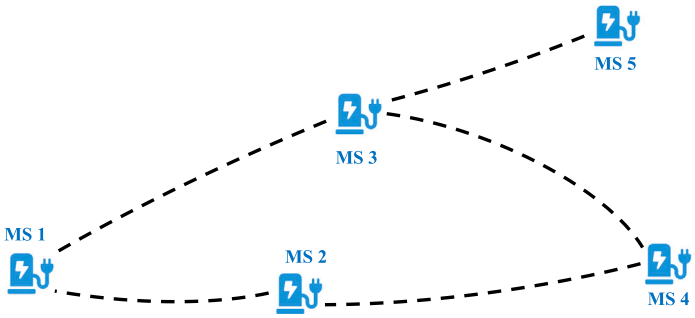
The performance of DCMGs is first analysed to illustrate the effect of workload migration. As shown in Fig. 6, workload migration enables flexible redistribution of critical computing tasks among the three DCMGs while maintaining zero workload curtailment. Compared with the case without migration, DCMG 1 and DCMG 2 transfer parts of their workloads to DCMG 3, resulting in net workload reductions of 57.24 and 582.76, respectively, over the scheduling horizon. Accordingly, DCMG 3 receives an additional workload of 640.00. This redistribution reshapes the spatial electricity demand of the DCMGs. In particular, the total electrical demands of DCMG 1 and DCMG 2 are reduced by 45.29 kWh and 465.27 kWh, respectively, whereas that of DCMG 3 increases by 206.96 kWh. Consequently, the aggregate electrical demand of the three DCMGs is reduced by 303.60 kWh, because workloads are shifted from the less energy-efficient DCMG 2 to the more energy-efficient DCMG 3, thereby reducing both server and cooling energy consumption while preserving the total workload. These results demonstrate that DCMGs are not merely passive critical loads but can actively support post-disaster restoration by shifting computing demand away from stressed locations.

The scheduling behaviours of the MPSs further demonstrate their spatial–temporal flexibility. As shown in Fig. 7, the MESS provides immediate power support by discharging 100 kW at hour 0. It subsequently recharges at hours 2, 4, 5, and 13 when the system conditions permit, and discharges again at hour 7 and during the evening periods, particularly during hours 17–20 and 22–23. Over the entire restoration horizon, the MESS discharges 504.00 kWh and charges 400.00 kWh. Meanwhile, the MEG provides power support throughout most of the restoration horizon and is operated at its maximum output of 150 kW during several critical periods, particularly during the evening peak. Its total energy production reaches 2629.11 kWh. The routing results further show that the MESS is relocated from station 3 to stations 2 and 4, while the MEG is successively relocated from station 3 to stations 2 and 4. These results indicate that the MPSs are dynamically deployed according to the changing spatial distribution of electricity demand and network conditions rather than remaining at fixed locations.

At the system level, the coordinated operation of DCMGs and MPSs improves the overall restoration performance, as illustrated in Fig. 8. When workload migration is considered, the total conventional unserved energy is reduced from 3586.85 kWh to 3476.53 kWh, corresponding to a reduction of 3.08%. Although the percentage improvement is moderate, it is achieved without installing additional generation, storage, or network capacity, since workload migration only redistributes flexible computing demand among the DCMGs. Moreover, this improvement represents an incremental benefit after network reconfiguration and the dispatch of the available DGs, MEGs, and MESSs have already been optimised. The system-level percentage is also limited by the large proportion of conventional demand that cannot be spatially transferred. Therefore, the reduction demonstrates that workload migration provides meaningful complementary flexibility by reducing both the cumulative unserved energy and the peak-period restoration pressure under resource-constrained conditions. More specifically, DCMG workload migration alleviates local demand stress, while MPS routing and scheduling provide power support to outage areas. Their complementary flexibility



(a) Modified IEEE 33-bus distribution network.



(b) Detailed road network including 5 MS stations.

Fig. 3. The investigated modified IEEE 33-bus distribution network with 3 DCMGs and 5 MS stations.

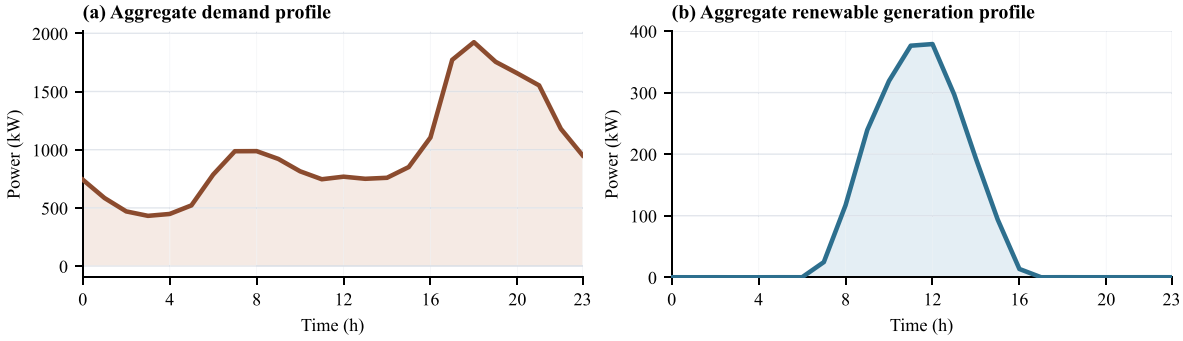


Fig. 4. Input data on aggregated demand and RESs in the modified IEEE 33-bus distribution network.

Table 1
Technical and thermal parameters of the DCs.

DCMG	Bus	Servers	D_i^{max} (h)	T_i^{in} range (°C)	Cooling cap. (kW)	A_i (m^2)	R_i (m^2 °C/kW)	C_i (kWh/°C)	η_i^h/η_i^c
DC 1	B3	60	0.08	18–24.8	19.04	500	750	5.5	1.0/3.0
DC 2	B14	60	0.08	18–24.8	30.60	500	750	5.5	1.0/3.0
DC 3	B21	60	0.08	18–24.8	20.40	500	750	5.5	1.0/3.0

assists the distribution network in maintaining critical service continuity and enhances resilience during the post-disaster restoration process.

To analyse the influence of outage duration on system restoration performance, a sensitivity analysis is conducted considering three outage durations, i.e., 12, 18, and 24 hours, while all other settings remain unchanged. The results are presented in Table 4. As the outage duration increases, the load shedding cost increases. In particular, compared with the 24-hour case, the 12-hour outage reduces the load shedding cost by 7.24%, demonstrating the adverse effect of prolonged outages on system restoration. As shown in Fig. 9, the distribution network is reconfigured into several islands after the extreme event. In particular, the tie lines B25–B29, B9–B15, B12–B22, and B18–B33 are closed for load

Table 2
Server power consumption and service-rate parameters.

DCMG	State	$P_{i,s}^{\text{fix}}$ (kW)	$k_{i,s}$ (kW/GHz^3)	$f_{i,s}$ (GHz)	$\mu_{i,s}$
DC 1	Low	0.700	0.175	1.000	3
DC 1	Mid	0.840	0.175	1.260	5
DC 1	High	0.980	0.175	1.533	8
DC 2	Low	1.440	0.360	1.000	3
DC 2	Mid	1.728	0.360	1.260	5
DC 2	High	2.016	0.360	1.533	8
DC 3	Low	0.660	0.165	1.000	3
DC 3	Mid	0.792	0.165	1.260	5
DC 3	High	0.924	0.165	1.533	8

restoration while maintaining the radial topology. The resulting reconfiguration bypasses the unavailable branches and enables the available black-start resources to support load restoration in the corresponding islands.

6.2. Results analysis of uncertainty modelling

Fig. 10 presents the asymmetric uncertainty envelopes and the corresponding worst-case realisations for PV generation, demand, and ambient temperature. Unlike a conventional symmetric uncertainty set that uses the same deviation margin in both directions, the proposed uncertainty set employs different upper and lower bounds to capture the direction-dependent characteristics of forecast errors. In this case, the calibrated relative deviation bounds are $\Delta^{\text{pv},+} = 5.08\%$ and $\Delta^{\text{pv},-} = 7.09\%$ for PV generation, $\Delta^{\text{ld},+} = 5.27\%$ and $\Delta^{\text{ld},-} = 3.26\%$ for demand, and $\Delta^{\text{tp},+} = 0.324\%$ and $\Delta^{\text{tp},-} = 0.171\%$ for ambient temperature.

These asymmetric margins indicate that the uncertainty characteristics vary across both channels and deviation directions. For PV generation, the larger downward margin and the corresponding worst-case trajectory show that renewable generation shortage is more critical for the restoration problem. In contrast, the load realisation tends to approach the upper boundary in several periods, increasing the supply burden of the damaged distribution network. For ambient temperature, the larger upward margin reflects the potential increase in cooling demand and its impact on the thermal dynamics of DCMGs. Overall, the worst-case trajectories do not expand uniformly around the nominal profiles, but selectively approach the upper or lower boundaries according to the adverse impact of each uncertainty channel. Moreover, the cardinality budgets prevent all uncertain parameters from simultaneously reaching their worst-case values over the entire horizon, thereby capturing adverse yet not overly conservative uncertainty realisations.

Fig. 11 further illustrates the data-driven construction of the asymmetric uncertainty bounds based on historical forecast errors. The empirical error distributions of PV generation, demand, and ambient temperature are not perfectly symmetric and exhibit different spreads in the positive and negative directions. This observation justifies the use of channel-specific and direction-dependent uncertainty margins rather than a single symmetric bound. The obtained bounds are wider for channels and directions with larger historical deviations, while remaining tighter for relatively stable or less critical deviations. By separately characterising positive and negative forecast errors, the proposed risk-tunable uncertainty modelling approach better preserves the statistical features of historical data and avoids unnecessary conservativeness caused by over-enlarging the less critical side of the uncertainty interval. As a result, the model can hedge against realistic adverse deviations while maintaining a less conservative feasible region for post-disaster restoration decisions.

To evaluate the price and benefit of robustness, the deterministic and robust solutions are compared under uncertainty budgets of $\Gamma = 6, 8, 10, 12$. The deterministic first-stage solution is first obtained using the nominal forecasts and remains fixed for all uncertainty budgets, whereas the robust first-stage solution is obtained separately for each value of Γ. First, both solutions are evaluated under the nominal forecasts to quantify the price of robustness. They are then evaluated under the same worst-case uncertainty realisation identified for the corresponding robust solution to quantify the resulting robustness benefit. In both evaluations, the respective first-stage decisions are fixed, while the second-stage continuous recourse variables are re-optimised under the considered worst-case scenario. The results for load shedding costs are summarised in Tables 5 and 6, while the differences in the first-stage decisions are reported in Table 7.

As shown in Tables 5 and 6, increasing the uncertainty budget raises the nominal price of robustness from 0.27% to 2.03%, while improving the objective under the common worst-case realisation by 0.95%–3.10%. This indicates that the robust solution achieves better worst-case restoration performance at the expense of a modest reduction in nominal economic performance. Moreover, Table 7 shows that the MESS routing, MEG routing, and switch decisions differ between the two solutions, confirming that the robust model proactively adjusts both MPS deployment and network reconfiguration in response to uncertainty.

Table 3
Technical parameters of mobile power sources.

Resource	Number	Initial Location	Candidate buses	$P^{\max}$(kW)	$Q^{\max}$(kVar)	$E^{\max}$(kWh)	Efficiency(%)
MEG	1	B14	B1,B6,B31,B14,B24	150	100	–	–
MESS	1	B24	B1,B6,B31,B14,B24	100	–	400	90

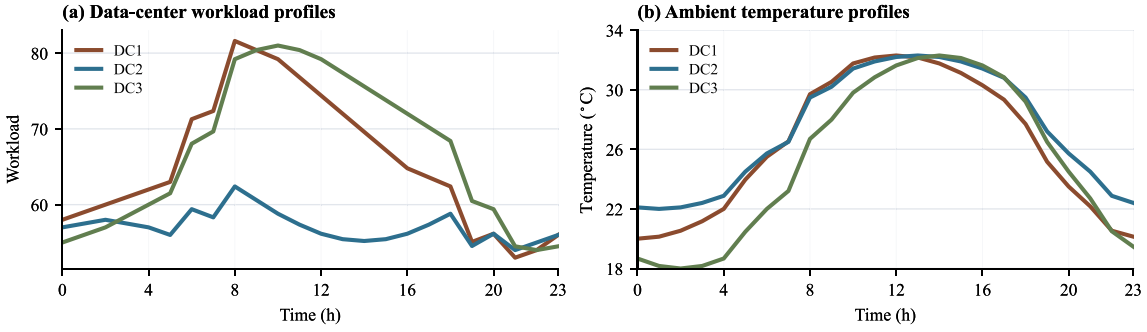


Fig. 5. Workload profiles and ambient temperature profiles of three DCs in the IEEE 33-bus distribution network.

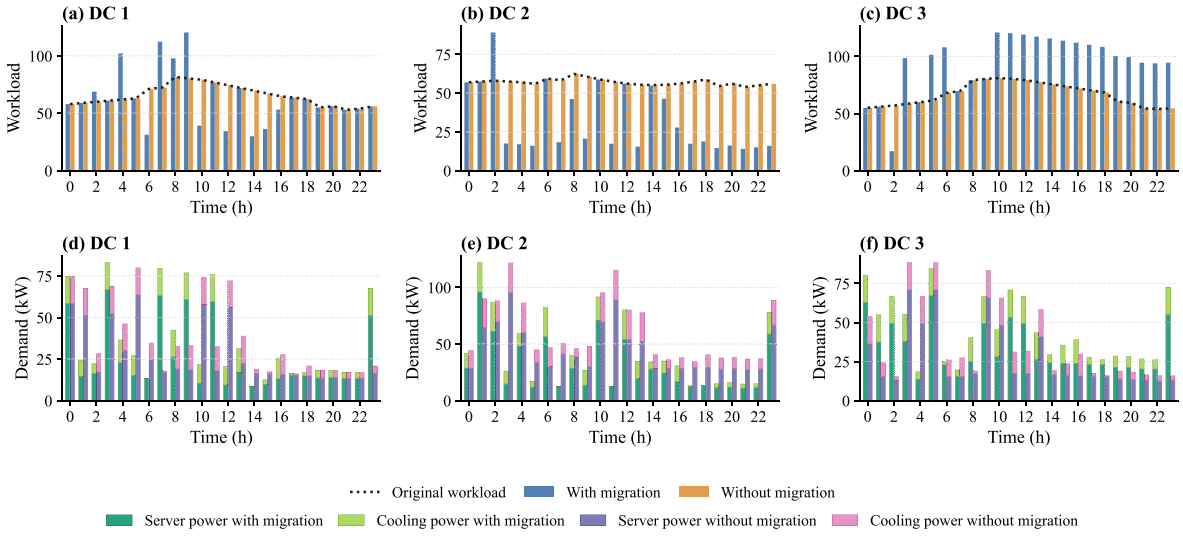


Fig. 6. Workload migration and corresponding demand profiles in three DCMGs on a representative day.

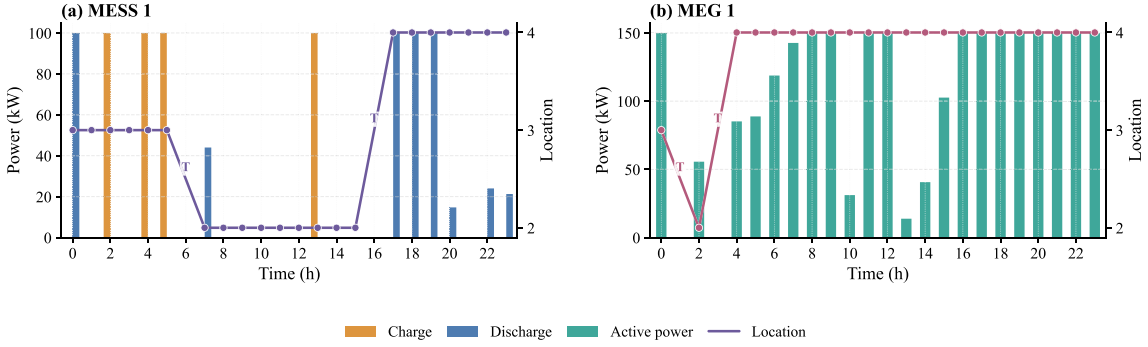


Fig. 7. Routing and scheduling behaviours of MESS and MEG in the distribution network.

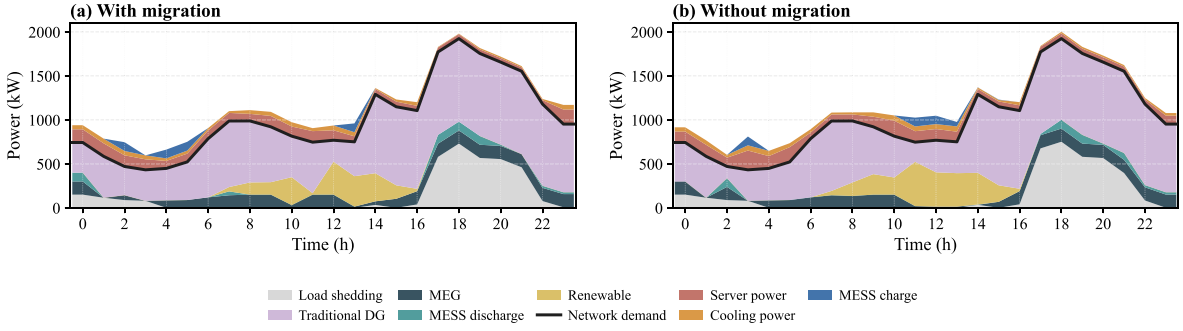


Fig. 8. Demand–supply balance in the case 'with workload migration' and the case 'without workload migration'.

Table 4
Sensitivity of restoration performance to the assumed outage duration.

Outage duration (h)	Interruption cost (£)	Cost reduction (%)
12	5780.61	7.24
18	6016.31	3.46
24	6231.74	0.00

6.3. Performance and sensitivity analyses

Two complementary analyses are conducted to evaluate the proposed framework: a component-wise performance analysis and a multiple-fault sensitivity analysis. In the first analysis, C1 employs symmetric uncertainty with the full recourse model, C2 employs asymmetric uncertainty with the full recourse model, and C3 employs asymmetric uncertainty with the topology-cleaned recourse model. Accordingly, the comparison between C1 and C2 evaluates the effect of asymmetric uncertainty characterisation, whereas the comparison between C2 and C3 evaluates the effect of the topology-cleaned recourse reformulation. In the second analysis, five outage scenarios are considered. Specifically, S1 contains one downstream fault, S2 contains two spatially dispersed faults, S3 contains three DC-adjacent faults, S4 contains four faults located in upstream and branch-root sections, and S5 contains five spatially dispersed faults. The scenarios therefore represent

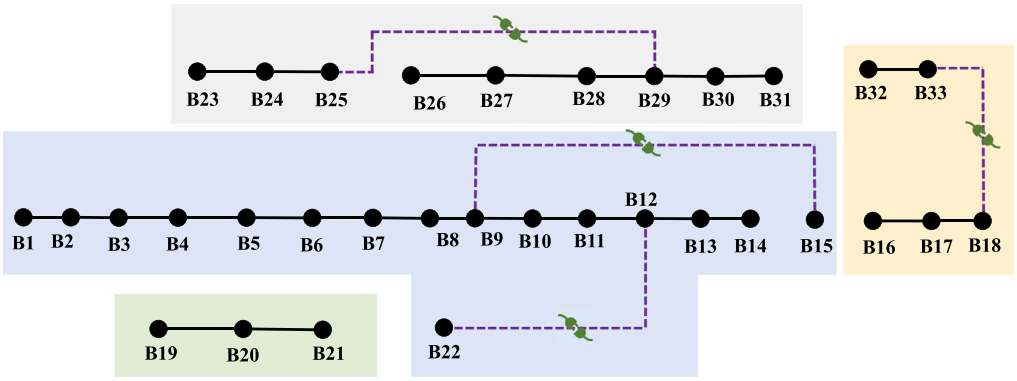


Fig. 9. The final distribution network reconfiguration strategy.

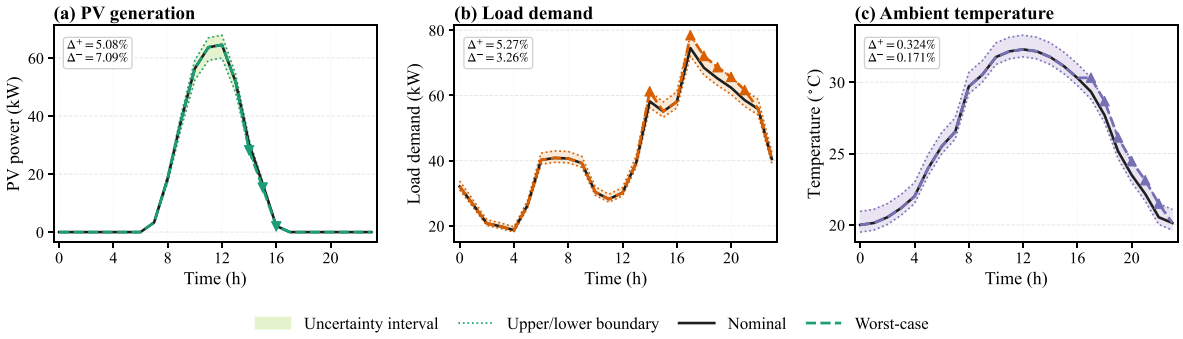


Fig. 10. Asymmetric uncertainty envelopes and worst-case realisations for the IEEE 33-bus distribution network: a) PV unit 1, b) load unit 1, and c) ambient temperature.

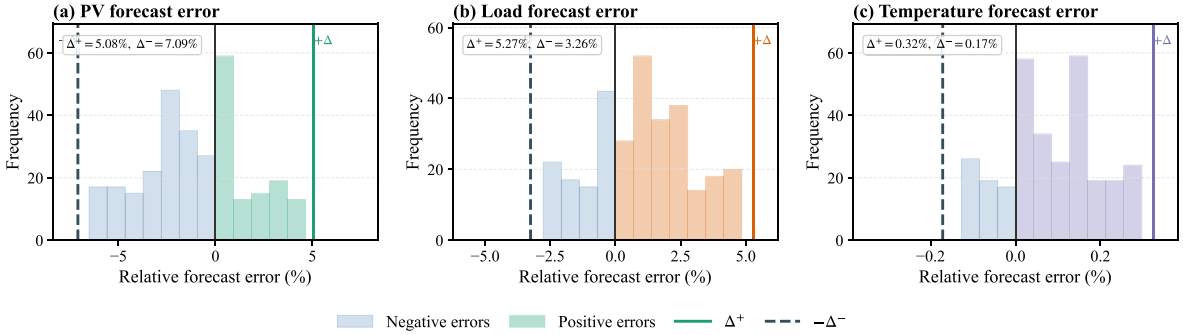


Fig. 11. Historical forecast-error distributions and data-driven asymmetric uncertainty bounds for the IEEE 33-bus distribution network: a) PV forecast error, b) load forecast error, and c) temperature forecast error.

Table 5
Nominal performance and price of robustness under different uncertainty budgets.

Uncertainty budget Γ	Deterministic objective (£)	Robust-plan objective (£)	Price of robustness (%)
6	5425.70	5440.17	0.27
8	5425.70	5495.26	1.28
10	5425.70	5521.31	1.76
12	5425.70	5535.75	2.03

Table 6
Load shedding costs under the same worst-case realisation.

Uncertainty budget Γ	Deterministic cost (£)	Robust cost (£)	Improvement (%)
6	6291.38	6231.74	0.95
8	6448.93	6352.63	1.49
10	6570.29	6433.64	2.08
12	6657.57	6451.28	3.10

different fault locations, numbers of failed components, damage severities, and initial islanding configurations. All cases use the same input data, uncertainty budgets, solver settings, and convergence tolerance.

As shown in Table 8, replacing the symmetric uncertainty representation in C1 with asymmetric uncertainty in C2 increases the worst-case objective slightly from £6192.64 to £6231.74, corresponding to an increase of approximately 0.63%. This occurs because the separately calibrated asymmetric bounds assign slightly larger deviations to the operationally adverse directions, including PV generation reduction, demand increase, and ambient-temperature increase. These direction-dependent

Table 7
First-stage differences between the deterministic and robust solutions.

Γ	MESS routing	MEG routing	Line status
6	9	5	17
8	10	8	16
10	9	8	16
12	7	7	16

deviations create a more demanding worst-case realisation than that obtained using the common symmetric bounds. C1 and C2 also produce different MESS and MEG routing decisions, demonstrating that the uncertainty characterisation affects the first-stage resource-allocation strategy. In comparison, C2 and C3 obtain identical final bounds, objective values, and first-stage decisions. Nevertheless, the topology-cleaned recourse reformulation reduces the number of constraints in the slave problem from 18,895 to 16,735, corresponding to a reduction of 11.43%, and decreases the computational time from 90.14 s to 82.32 s, leading to a reduction of 8.67%. The unchanged master-problem size confirms that the reduction is achieved in the slave/recourse problem rather than in the outer C&CG master problem. All three cases terminate with a reported C&CG optimality gap of 0.000%, satisfying the prescribed convergence tolerance.

Table 9 shows that the proposed framework obtains converged restoration strategies for all five fault scenarios within two or three iterations, with restored-load percentages ranging from 79.46% to 85.84%. S3 produces the highest load shedding cost and lowest restored-load percentage because its faults are adjacent to the DCMGs, although it does not contain the largest number of failed components. By contrast, S4 has the highest initially disconnected-load percentage but maintains 85.05% load restoration through coordinated network reconfiguration, local generation, and mobile resources. The different MESS and MEG movements across the scenarios further show that the mobile resources adapt to the spatial distribution of the faults. These results indicate that restoration performance and computational difficulty depend on fault location and islanding structure, rather than only on the number of failed components.

6.4. Validation on larger distribution networks

This section further demonstrates the scalability and effectiveness of the proposed resilience-oriented coordination scheme on a larger distribution network, i.e., a modified IEEE 69-bus distribution system with 6 DCMGs. As shown in Fig. 12, the 6 DCMGs are allocated at different buses of the distribution network, while 4 MESSs and 4 MEGs are deployed as MPSs to provide mobile emergency support during post-disaster restoration. Compared with the smaller case study, this larger system involves more DCMGs and MPSs, and is therefore used to verify the effectiveness of the proposed resilience-oriented coordination framework in making realistic decisions under increased system complexity. The key results, including the coordinated operation overview, demand–supply balance, and worst-case uncertainty realisation, are demonstrated in Figs. 13–15.

Fig. 13 provides a compact overview of the coordinated behaviours of DCMGs and MPSs in the 69-bus network. The heatmap in Fig. 13(a) illustrates the workload redistribution among the 6 DCMGs, where positive values indicate that a DCMG processes additional migrated workload, while negative values indicate that part of its original workload is transferred to other DCMGs. It can be observed that workload migration is activated across multiple DCMGs and time periods rather than being limited to a single DC, demonstrating the spatial flexibility of DCMGs in reshaping electricity demand during the restoration process. For instance, several DCMGs choose to reduce their workload in stressed periods, while others with more favourable operating conditions absorb additional workload, thereby supporting overall system-level load restoration. As exhibited in Fig. 13(b), MEGs are mainly dispatched during periods with high restoration pressure, whereas MESSs exhibit both charging and discharging behaviours, indicating their temporal flexibility in absorbing surplus energy and providing power support when needed. These coordinated behaviours verify that the proposed approach can jointly exploit workload migration, mobile generation, and mobile storage in a larger distribution network. The corresponding system-level demand–supply balance in Fig. 14 further demonstrates that RESs, DGs, MEGs, MESS discharge, and load shedding jointly contribute to meeting the total demand and the additional demand from DCMGs and MESS charging. Overall, Figs. 13 and 14 verify that the proposed coordination framework remains effective in managing multiple DCMGs and MPSs in a larger-scale distribution network.

The results of uncertainty modelling are illustrated in Fig. 15, where the uncertainty envelopes are asymmetric with the corresponding worst-case realisations for RES generation, demand, and ambient temperature. In other words, the worst-case trajectories do not simply deviate uniformly from the nominal forecasts; instead, they selectively approach the upper or lower bounds according to their operational impact on the restoration process. For RES generation, downward deviations are particularly critical, since they reduce available supply during restoration. For demand and ambient temperature, upward deviations are more adverse, since they increase conventional demand and cooling-related pressure on DCMGs. These results demonstrate that the proposed data-driven uncertainty characterisation can capture channel-specific and direction-specific risk features in a larger-scale distribution network; it is scalable to various complex networks with multiple DCMGs, MPSs, and uncertainties.

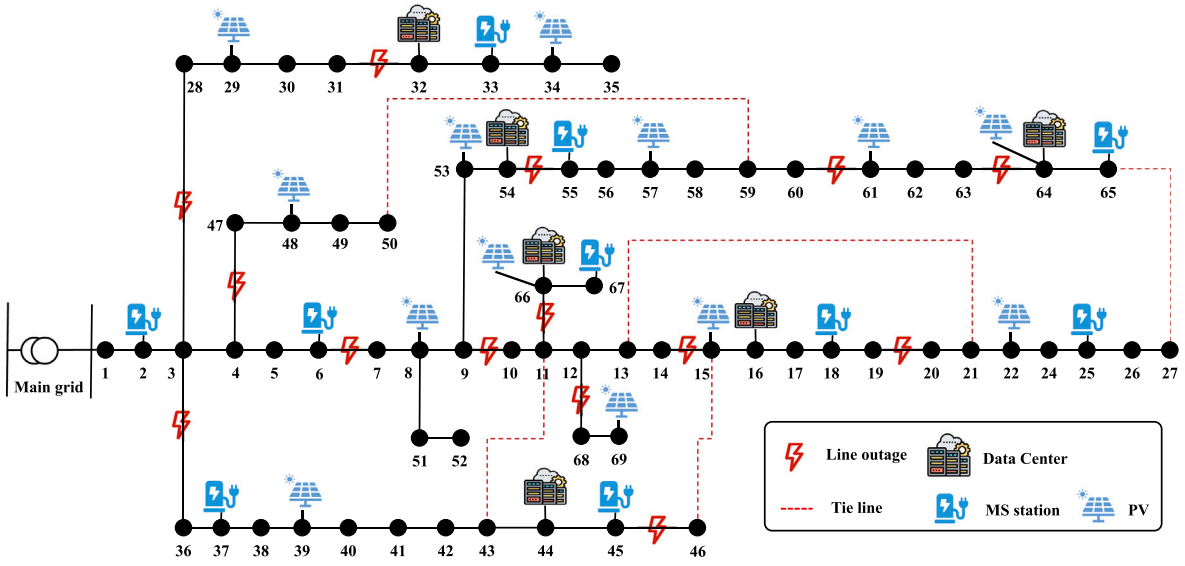
To further evaluate the applicability and scalability of the proposed framework, an additional case study is conducted on a modified IEEE

Table 8
Performance comparison of different uncertainty and recourse formulations.

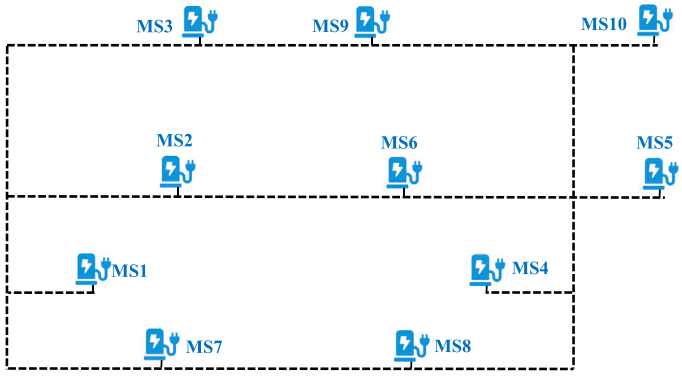
Case	Iter.	Time (s)	Final LB (£)	Final UB (£)	Final obj. (£)	Final MP size (vars./cons.)	SP constraints
C1	3	85.26	6192.64	6192.64	6192.64	44,059/104,067	18,895
C2	3	90.14	6231.74	6231.74	6231.74	44,059/104,067	18,895
C3	3	82.32	6231.74	6231.74	6231.74	44,059/104,067	16,735

Table 9
Sensitivity analysis under multiple fault scenarios in the 33-bus system.

Scenario	Initial islands	Disconnected load (%)	Objective (£)	Restored load (%)	Iter.	Time (s)	MESS/MEG moves
S1	2	8.73	5270.71	85.84	2	57.78	2/5
S2	3	39.75	5520.33	85.17	2	140.44	5/3
S3	4	87.27	7649.23	79.46	2	438.63	4/3
S4	5	95.76	5849.45	85.05	2	378.43	3/2
S5	6	52.33	6231.74	83.83	3	224.57	4/5



(a) Modified IEEE 69-bus distribution network.



(b) Road network including 10 MS stations.

Fig. 12. Illustration of the investigated IEEE 69-bus distribution network with 6 DCMGs and 10 MS stations.

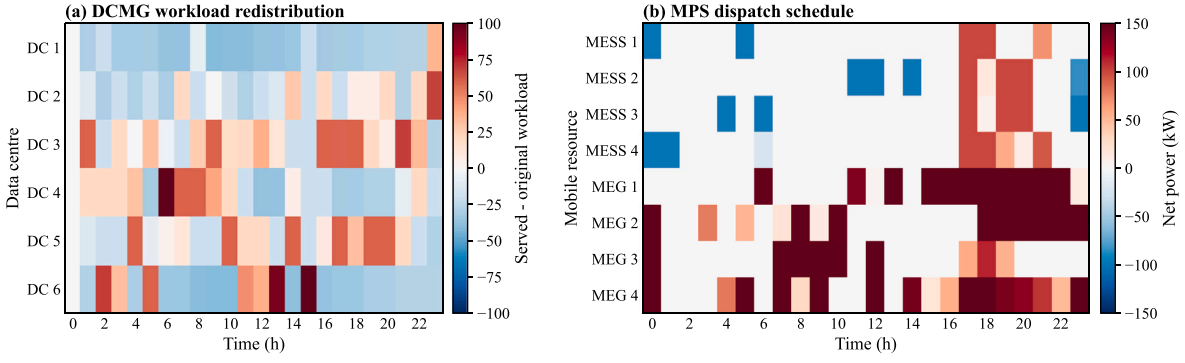


Fig. 13. Illustration of DCMG workload redistribution and MPS dispatch schedule in the IEEE 69-bus distribution network.

123-bus distribution network, which contains 8 MESSs, 8 MEGs, and 12 DCMGs. All cases use identical uncertainty budgets and the same C&CG convergence tolerance. The system configurations and computational performance of the 33-bus, 69-bus, and 123-bus systems are summarised in Tables 10 and 11, while the influence of the scheduling horizon is further reported in Table 12. As summarised in Table 10, the modified 123-bus distribution network contains 3744 uncertain parameter entries, 118,776 variables, and 208,844 constraints.

It can be observed from Table 11 that the C&CG algorithm for the 123-bus case converges in three iterations to an objective value of £24,965.45, with a reported C&CG optimality gap of 0.000% at the adopted numerical precision. The total solution time is 1090.2 s and the peak memory requirement is 3830.0 MB, where the per-iteration time increases from 27.4 s for the 33-bus case to 201.0 s for the 69-bus case and 363.4 s for the 123-bus case. Table 12 further shows that the number of uncertain parameter entries, variables, and constraints

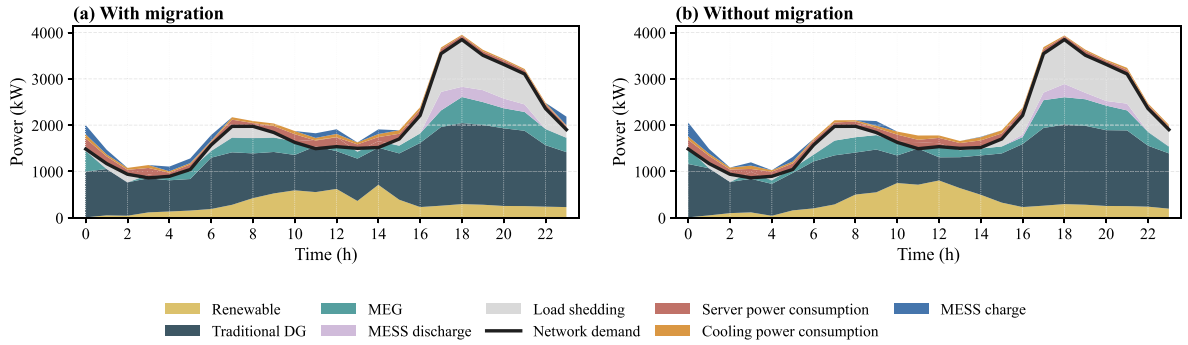


Fig. 14. Demand–supply balance for the case 'with workload migration' and the case 'without workload migration' in the IEEE 69-bus distribution network.

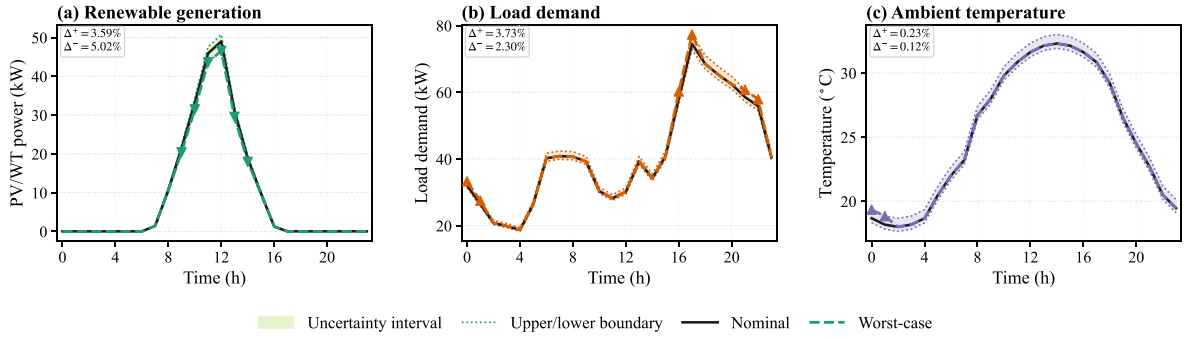


Fig. 15. Asymmetric uncertainty envelopes and worst-case realisations for the IEEE 69-bus distribution network: a) PV unit 1, b) load unit 1, and c) ambient temperature.

Table 10
Configuration and nominal-model complexity for different network sizes.

System	Loads	PV	DCMGs	MESS/MEG	Uncertain parameters	Variables	Binary variables	Constraints
33-bus	21	6	3	1/1	720	14,784	3360	32,448
69-bus	42	12	6	4/4	1440	47,952	11,664	113,536
123-bus	120	24	12	8/8	3744	118,776	17,760	208,844

Table 11
C&CG performance and final problem sizes for different network sizes.

System	Iter.	Computing time (s)	Objective (£)	Gap (%)	Peak memory (MB)	Master variables/constraints	Slave variables/constraints
33-bus	3	82.3/27.4	6231.74	0.000	727.2	44,059/104,067	2712/16,735
69-bus	4	804.0/201.0	19,068.62	0.000	2125.1	189,841/489,140	5760/41,797
123-bus	3	1090.2/363.4	24,965.45	0.000	3830.0	349,105/662,055	14,304/90,565

Table 12
Scheduling-horizon scaling of the deterministic model for the modified IEEE 123-bus case.

Periods	Uncertain parameters	Variables	Binary variables	Constraints	Nominal solve time (s)
12	1872	59,388	8880	104,408	26.4
18	2808	89,082	13,320	156,626	51.5
24	3744	118,776	17,760	208,844	61.2

increases approximately linearly with the scheduling horizon. All three deterministic models are solved to optimality. These results demonstrate that the proposed formulation remains computationally tractable for a distribution network with more than 100 buses, while also quantifying the additional time and memory required by the enlarged recourse and uncertainty spaces.

7. Conclusions

This paper develops a resilience-oriented coordination framework for DCMGs and MPSs to enhance the resilience of power distribution networks under extreme events. In the proposed framework, DCMGs actively manage their electricity demand through workload migration, thereby reducing the load restoration pressure in outage-affected areas. Meanwhile, MPSs support post-disaster service restoration through effective routing and scheduling strategies. The coordinated operation of DCMGs and MPSs enables a higher level of resilience by jointly exploiting demand-side flexibility and mobile supply-side resources. To address uncertainties associated with RESs, demand, and outdoor temperature, a data-driven two-stage distributionally robust optimisation model is proposed based on asymmetric uncertainty sets and Wasserstein distance. The proposed uncertainty characterisation can capture the different operational impacts of upward and downward deviations of uncertain inputs, such as RES reduction, demand increase, and temperature rise.

The two-stage model is reformulated into a master-slave structure and solved using a problem-tailored CC&G algorithm, which ensures stable convergence while progressively identifying critical uncertainty scenarios. Case studies are carried out on two distribution networks with different numbers of DCMGs and MPSs. The results verify the superiority of the proposed method in resilience enhancement, resource coordination, and uncertainty handling. Specifically, by considering DCMGs and their workload migration capability load shedding can be effectively reduced and overall system resilience. In addition, the introduced asymmetric uncertainty set provides a more realistic and reliable uncertainty modelling framework by distinguishing the specific impacts of different uncertainty sources and deviation directions, leading to more convincing and robust coordination decisions. Future work will investigate the influence of more detailed degradation characteristics, including depth of discharge, battery temperature, and state-of-charge-dependent ageing, on long-term multi-period resilience scheduling.

CRediT authorship contribution statement

Yi Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Liangzhong Yao:** Writing – review & editing, Validation, Resources. **Goran Strbac:** Writing – review & editing, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Linearisation of quadratic constraints

The circular feasible regions defined by the apparent-power constraints (29), (31), and (35) are approximated by N-segment inscribed polygons [49]. Taking the line-flow constraint (35) as an example, it can be replaced by the following set of linear inequalities:

$$P_{bp,t}\cos\theta_n + Q_{bp,t}\sin\theta_n \le y_{bp,t}\overline{S}_{bp}\cos\left(\frac{\pi}{N}\right), \forall n \in \mathcal{N} \tag{A.1}$$

where

$$\theta_n = \frac{(2n-1)\pi}{N}, \mathcal{N} = \{1,\dots,N\} \tag{A.2}$$

The PV apparent-power constraints (29) and (31) are linearised in the same manner. Since the polygon is inscribed within the original circular region, the approximation is conservative. After the first-stage binary variables are fixed, the resulting second-stage constraints are linear, thereby preserving the LP structure required by the dual reformulation.

Data availability

No data were used for the research described in the article.

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